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APPENDIX A: PRESERVED AUTHORITIES AND PROCEDURAL PREDICATES

This Appendix preserves for the record arguments, authorities, and procedural predicates that Petitioner has developed in the course of this proceeding but which, in the interest of brevity and focus, are not fully briefed in the body of this Reply. Nothing in this Appendix is offered as a comprehensive legal brief. Each entry identifies the argument, the controlling or persuasive authority, and the record basis for its preservation. All arguments preserved herein are independently maintained for petition for reconsideration under 47 U.S.C. Section 405, application for review under 47 C.F.R. Section 1.115, and judicial review under 47 U.S.C. Section 402(b) and 5 U.S.C. Section 706. To the extent any argument identified below has been more fully developed in any prior filing in this docket, that development is incorporated by reference and controls.

I. DELEGATED AUTHORITY AND THE NOVEL QUESTIONS REFERRAL OBLIGATION

Argument Preserved: The Space Bureau's delegated authority under 47 C.F.R. Section 0.51 does not encompass first-impression questions about the Commission's regulatory jurisdiction, the adequacy of its environmental review framework for systems of unprecedented scale, or the sufficiency of an ownership disclosure leaving the majority of equity unattributed. Under the Commission's own delegation framework, matters presenting novel questions of law, fact, or policy that cannot be resolved under existing precedents and guidelines, should be referred to the Commission en banc for disposition under the Commission's own delegation framework, which reserves matters presenting novel questions of law, fact, or policy for full Commission deliberation. See 47 C.F.R. Section 0.283(c) (Media Bureau model, applicable by structural analogy to the Space Bureau).

Supporting Authorities:

Commissioner Pai, "Delegated Authority" (February 2015): "[T]here are certain aspects of the FCC's duties that should be reserved and addressed by the full Commission. Chief among those are matters that are new or novel."

Commissioner Pai, "Modified Delegated Authority" (February 2017): Proposed 48-hour standardization of advance Commissioner notification before Bureau-level action on delegated items.

47 C.F.R. Section 0.331 (delegation structure); 47 U.S.C. Section 155(c)(4) (Commission retains authority to review any action taken under delegated authority).

47 C.F.R. Section 1.117 (actions taken under delegated authority subject to review by the Commission on its own motion or on application for review).

Record Basis: The Commission's own October 2025 Part 100 NPRM (FCC 25-69, SB Docket No. 25-306) acknowledges that the existing Part 25 framework presents novel questions when applied to systems of this type, proposing an entirely new regulatory framework to address them. The fact that the Commission is simultaneously proposing new rules to govern novel satellite systems while processing this Application under the old rules is itself a basis for requiring full Commission deliberation before final action. At least four independent novel questions are presented: (1) whether Part 25 authorizes licensing of a system whose primary function is orbital AI computing; (2) whether an environmental certification can be sustained on a record containing no aggregate atmospheric data against peer-reviewed science; (3) whether the cumulative waiver package can be evaluated under existing precedent; and (4) whether the ownership disclosure satisfies mandatory requirements where the post-merger ownership picture is not before the Commission.

II. THE GOVERNMENT IN THE SUNSHINE ACT AND NOTATION VOTING LIMITS

Argument Preserved: If any disposition of this Application involves the deliberation of a quorum of Commissioners, the Government in the Sunshine Act, 5 U.S.C. Section 552b, requires that such deliberation be conducted in public session unless a specific statutory exemption is invoked and voted upon. Notation voting is not an adequate substitute for open deliberation where the matter is non-routine.

Supporting Authorities:

5 U.S.C. Section 552b(b): "[E]very portion of every meeting of an agency shall be open to public observation."

5 U.S.C. Section 552b(a)(2): Defines "meeting" as the deliberations of at least the number of individual agency members required to take action on behalf of the agency where such deliberations determine or result in the joint conduct or disposition of official agency business.

Communications Systems, Inc. v. FCC, 595 F.2d 797 (D.C. Cir. 1978): Held that notation voting does not violate the Sunshine Act for routine matters. The operative qualifier is "routine."

McChesney v. FEC, 900 F.3d 578 (8th Cir. 2018): Confirmed that notational voting procedure in a routine matter did not violate the Sunshine Act. The holding is limited to routine matters.

5 U.S.C. Section 552b(d)(1): Requires a separate recorded vote of agency members with respect to each meeting proposed to be closed. No proxies.

5 U.S.C. Section 552b(e)(1): Requires public announcement at least one week before the meeting of the time, place, and subject matter.

Record Basis: The proceeding before the Commission is not routine by any measure. The Chairman has made public statements about the Application on X (February 4 and March 11, 2026). The Applicant formally cited the March 11 statement at footnote 5 of its Consolidated Opposition as supporting authority. Commissioner engagement with the Application's branding during the pleading cycle has been documented on the record. See Bakken IR, Request 4 and Request 7. Where individual Commissioners have publicly expressed views about a pending adjudicatory matter during the active pleading window, the deliberation the Sunshine Act is designed to make visible raises concerns that the Act was designed to address, particularly where the Applicant has cited those expressions as supporting authority.

The Act was designed to prevent precisely that condition.

III. SECTION 309(e) MANDATORY HEARING DESIGNATION

Argument Preserved: Where the Commission is unable to make the affirmative public interest finding required by 47 U.S.C. Section 309(a), or where a party in interest raises a substantial and material question of fact, Section 309(e) commands hearing designation. The statute uses "shall." This is not discretionary.

Supporting Authorities:

47 U.S.C. Section 309(e): The Commission "shall formally designate the application for hearing" where it cannot make the Section 309(a) finding or where a substantial and material question of fact is presented.

RKO General, Inc. v. FCC, 670 F.2d 215 (D.C. Cir. 1981)

Citizens for Jazz on WRVR, Inc. v. FCC, 775 F.2d 392 (D.C. Cir. 1985)

DA 26-76 (January 23, 2026): 97.5 Licensee TX, LLC. The most recent invocation of the Commission's hearing designation authority. Character and candor policy applied.

102 F.C.C.2d 1179 (1986): FCC character and candor policy. Misrepresentation and lack of candor are grounds for hearing designation independent of ultimate merits. See also Contemporary Media, Inc. v. FCC, 214 F.3d 187 (D.C. Cir. 2000).

Record Basis: Seven independent formal parties have placed factual disputes on this record. The certified technical data is mathematically irreconcilable with the narrative. The Ownership Attachment is materially stale. The environmental certification is contradicted by peer-reviewed scientific evidence. The Opposition did not cure any of these deficiencies. These are specific, documented factual disputes on the record, not generalized policy disagreements. The Schedule S encodes 3 satellites; the Narrative describes 1,000,000. The Ownership Attachment predates the merger. The environmental certification is contradicted by the Applicant's own phased deployment admission at Opposition page 16. Each is a concrete factual inconsistency the

Commission must resolve, and the statute provides the mechanism: hearing designation under Section 309(e).

IV. MAJOR QUESTIONS DOCTRINE

Argument Preserved: Licensing a system whose declared primary purpose is orbital AI computing infrastructure under a statute enacted to govern radio communications presents a question of vast economic and political significance for which the Commission must point to clear congressional authorization.

Supporting Authorities:

West Virginia v. EPA, 597 U.S. 697 (2022). The Court held 6-3 that the EPA exceeded its authority under the Clean Air Act when it used a general provision to restructure the national electricity grid through "generation shifting," moving production from coal to cleaner sources. The Court held that where an agency claims authority to resolve a question of "vast economic and political significance," it must point to "clear congressional authorization" for the power it claims. The Court found the EPA had discovered an "unheralded power" in a statutory provision it had used only sparingly for decades, to justify a regulatory scheme of enormous economic consequence that Congress had not clearly authorized. The Gorsuch concurrence identified a three-factor framework for recognizing major questions cases: (1) the legislative provisions in question, rejecting broad assertions from vague or cryptic language; (2) the age and focus of the statute in relation to the problem addressed; and (3) the agency's past interpretation of the relevant statute.

Biden v. Nebraska, 600 U.S. 477 (2023). The Court held 6-3 that the Secretary of Education lacked authority to cancel approximately \$430 billion in federal student loans under the HEROES Act, a post-September 11 statute authorizing the Secretary to "waive or modify" student loan provisions during a national emergency. The Court held that the word "modify" carries a connotation of incremental or limited change and cannot be stretched to authorize the wholesale transformation of a loan program. Applying the major questions doctrine as reinforcement, the Court found the economic significance "staggering by any measure," that Congress had considered and rejected broad student loan cancellation, and that no prior Secretary had asserted such sweeping authority under the statute. The Court held that the major questions doctrine applies even when the issuing agency possesses relevant subject-matter expertise, and that an emergency statute does not receive a broader reading simply because its purpose is to respond to emergencies.

Learning Resources, Inc. v. Trump, 607 U.S. ____ (2026). The Court held 6-3 that the International Emergency Economic Powers Act does not authorize the President to impose tariffs. The majority held that the power to "regulate ... importation" does not include the constitutionally distinct power to tax imports, and that where Congress has delegated tariff

authority elsewhere it has done so in explicit terms with strict limitations. A Roberts-Gorsuch-Barrett plurality applied the major questions doctrine, holding there is no exception for emergency statutes or foreign affairs, that no President had invoked IEEPA to impose tariffs in the statute's fifty-year history, and that the claimed authority to set tariffs "from any country, of any product, at any rate, for any amount of time" was precisely the kind of extraordinary assertion requiring clear congressional authorization. Justice Gorsuch's concurrence defended the major questions doctrine as a structural separation-of-powers safeguard against executive lawmaking through statutory gaps.

Record Basis: The Communications Act is nearly a century old. The satellite provisions date to 1962. Neither was designed to address orbital artificial intelligence infrastructure. The Commission has never licensed a system whose primary function is AI computing. The pending Part 100 NPRM (FCC 25-69) is itself evidence that the existing Part 25 framework was not designed to fully accommodate what this Application proposes.

Petitioner recognizes that the major questions doctrine has not yet been applied to an FCC licensing decision and preserves it here. The Supreme Court and lower courts have increasingly applied the doctrine across Executive Branch contexts where asserted authority carries vast economic and political significance and lacks clear congressional authorization. The trajectory is clear: from an agency restructuring the electricity grid (West Virginia), to an agency transforming a student loan program (*Biden v. Nebraska*), to a President claiming tariff authority from a general regulatory statute (*Learning Resources*). In each case, the claimed authority was "unheralded," the statute was not designed for the asserted purpose, and no prior exercise of such authority existed. Each of those conditions is present here. Part 25 governs "communication by radio." The Applicant's system primarily computes, not communicates. The Commission's own NPRM concedes the framework was not built for this. To the extent the Commission relies on general Part 25 authority to license a system whose primary function is orbital AI computing infrastructure at unprecedented scale, that reliance raises the same concerns regarding unheralded regulatory expansion that the Court identified in all three cases.

V. APA REASONED DECISION-MAKING, WHOLE-RECORD REVIEW, AND SECTION 555(e)

Argument Preserved: A grant on the current record would face challenge under the arbitrary and capricious standard on multiple independent grounds. To the extent the Commission denies, declines to act upon, or silently resolves any of the Bakken IR's fourteen requests without stated grounds, Section 555(e) is independently triggered for each such disposition.

Supporting Authorities:

Motor Vehicle Manufacturers Association v. State Farm Mutual Automobile Insurance Co., 463 U.S. 29 (1983).

Department of Commerce v. New York, 588 U.S. 752 (2019): An agency order whose stated reasons are not its actual reasons is arbitrary under the APA.

FCC v. Fox Television Stations, 556 U.S. 502 (2009) (Kennedy, J., concurring).

FCC v. Prometheus Radio Project, 141 S.Ct. 1150 (2021).

5 U.S.C. Section 555(e); Citizens to Preserve Overton Park v. Volpe, 401 U.S. 402 (1971).

5 U.S.C. Section 706(2)(A) and (C).

Record Basis: The Bakken IR consolidates fourteen independent requests, each triggering Section 555(e). Peer-reviewed science contradicting the environmental certification has been placed on the record and cannot be disregarded under Fox Television Stations. The Chairman's public statements during the pleading cycle, cited by the Applicant as supporting authority, raise the Department of Commerce v. New York question of whether the agency's stated reasons are its actual reasons.

VI. INADVERTENT ACCEPTANCE AND THE COMMISSION'S CONTINUING AUTHORITY

Argument Preserved: Acceptance of a defective application does not cure the defect. The Commission retains authority to return or dismiss an inadvertently accepted application at any time before grant.

Supporting Authorities:

47 C.F.R. Section 25.112(a); 47 C.F.R. Section 73.3564(b).

Alpha Broadcasting, Inc., Letter, DA-09-2464, 24 FCC Rcd 14068 (IB 2009).

Susquehanna Radio Corp. (KTDK(FM), Sanger, TX), Letter, DA-13-1938, 28 FCC Rcd 13662 (MB 2013).

Record Basis: The Application was accepted within approximately two business days during a government shutdown. See Bakken IR, Request 9. The record now contains documented deficiencies identified by seven independent formal parties.

VII. BLANKET WAIVER PROHIBITION AND THE CERTIFICATION PARADOX

Argument Preserved: The Commission's 2023 Final Rule explicitly prohibits blanket waivers. The Application's waiver narrative does not satisfy the specificity requirement. The simultaneous

certification of completeness and acknowledgment that the form cannot describe the system creates a material inconsistency.

Supporting Authorities:

FCC 23-73 (December 2023 Final Rule).

Form 312 Instructions (April 2024 edition, OMB Control Number 3060-0678).

47 C.F.R. Section 1.3.

DA-25-1018 (December 2025, SCS/Direct-to-Cell): SpaceX identified the specific rule by number in that proceeding. Demonstrates awareness of the specificity requirement.

Record Basis: The Applicant's own waiver narrative asserts that the filing vehicle cannot accurately describe the proposed system. That is a generalized position, not a specific rule-by-rule waiver request. The Applicant's prior practice in the SCS proceeding confirms awareness of the specificity requirement. The absence of that specificity in this Application is a choice, not an oversight.

VIII. MAJOR AMENDMENT CONSEQUENCES

Argument Preserved: Any supplementation the Applicant provides to cure the identified deficiencies would constitute a major amendment under 47 C.F.R. Section 25.116(b) on multiple independent grounds, triggering new public notice under Section 25.151 and, for NGSO applications, causing the application to be treated as newly filed under Section 25.116(c).

Supporting Authorities:

47 C.F.R. Section 25.116(b)(1), (b)(2), (b)(4); Section 25.116(c); Section 25.151(d).

Record Basis: Updated ownership information alters ownership characteristics. Bounded technical parameters at the described scale alter interference potential. Environmental baseline data responsive to the Bakken EP could convert the proposal into an action with significant environmental effect. Any one independently satisfies Section 25.116(b). The fork identified in the Bakken SIO remains operative: adjudicate on the record as filed, or acknowledge that any cure constitutes a material amendment requiring new public notice.

IX. CONTINUING ACCURACY OBLIGATION AND SECTION 1.65

Argument Preserved: The Applicant's obligation under 47 C.F.R. Section 1.65 to amend its application following a material change in ownership is self-executing, arises within thirty days, and continues until Commission action is no longer subject to reconsideration or judicial review. On the facts documented on this record, the conditions that trigger Section 1.65's self-executing amendment obligation appear to have been satisfied, and the record does not reflect that the obligation has been discharged. To the extent the Applicant has filed an amended disclosure

through channels not visible on the public docket, Petitioner respectfully requests that the Commission confirm that fact on the record.

Supporting Authorities:

47 C.F.R. Section 1.65; 47 U.S.C. Sections 310(a) and (b); Executive Order 13913; 47 C.F.R. Sections 1.5000 et seq.

Record Basis: The merger was completed February 2, 2026. The thirty-day period has expired. Formal notice was provided in this proceeding on March 6, 9, 15, 16, and 17. No amended filing has been submitted to date.

X. PREJUDGMENT AND THE SANGAMON VALLEY STANDARD

Argument Preserved: The Chairman's public statements during the active pleading cycle, and their citation by the Applicant as supporting authority, raise prejudgment concerns.

Supporting Authorities:

Sangamon Valley Television Corp. v. United States, 269 F.2d 221 (D.C. Cir. 1959).

Cinderella Career and Finishing Schools, Inc. v. FTC, 425 F.2d 583, 591 (D.C. Cir. 1970): "[A] disinterested observer may [not] conclude that [the agency] has in some measure adjudged the facts as well as the law of a particular case in advance of hearing it."

Record Basis: The Chairman posted about the Application on X on February 4, 2026, the same day the Public Notice was released. The Chairman posted again on March 11, 2026, publicly criticizing a statutory petitioner. The Applicant formally cited the March 11 statement at footnote 5 of its Opposition. These statements were made on a platform owned by the Applicant's corporate parent during the active pleading cycle. See Bakken IR, Request 7. Petitioner does not assert that these facts establish prejudgment as a matter of law. Petitioner asserts that these facts, taken together, raise the concerns the Sangamon Valley and Cinderella Career courts identified as warranting scrutiny, and that the Commission should address them on the record rather than allow them to persist as a basis for subsequent challenge.

XI. INTERNATIONAL TREATY OBLIGATIONS, ITU COORDINATION, AND THE APPLICANT'S OWN TREATY CITATION

Argument Preserved: The record does not contain ITU advance publication information, coordination data, or any indication of the status of international frequency coordination for the proposed system. The Applicant's own invocation of the Outer Space Treaty in its Opposition undermines rather than supports its extraterritorial argument. The United States' treaty obligations impose affirmative duties on the Commission that are directly relevant to this Application and that cannot be discharged on the current record.

Supporting Authorities:

A. ITU Radio Regulations

ITU Radio Regulations, Article 9 (Procedure for Effecting Coordination): Requires advance publication and coordination of frequency assignments for satellite systems before they may be brought into use. The purpose is to ensure that proposed systems do not cause unacceptable interference to the assignments of other administrations. The obligation runs to the notifying administration, which for U.S.-licensed systems is the United States acting through the FCC and NTIA.

ITU Radio Regulations, Article 11 (Notification and Recording of Frequency Assignments): Requires formal notification of frequency assignments to the Radiocommunication Bureau for recording in the Master International Frequency Register. Assignments not recorded do not enjoy international recognition or protection.

ITU Radio Regulations, Article 22 (Sharing Between NGSO Systems): Establishes equivalent power flux-density limits and coordination triggers for NGSO systems sharing spectrum with GSO and other NGSO networks. Compliance cannot be evaluated without bounded technical parameters at the proposed scale.

ITU Radio Regulations, Resolution 35 (Advance Publication and Coordination of Non-Geostationary Satellite Systems): Addresses the specific coordination procedures applicable to NGSO constellations, including the obligation to provide sufficient technical data to allow other administrations to assess interference potential.

B. The Outer Space Treaty (1967)

Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, Including the Moon and Other Celestial Bodies, Jan. 27, 1967, 18 U.S.T. 2410, 610 U.N.T.S. 205.

Article II: "Outer space, including the Moon and other celestial bodies, is not subject to national appropriation by claim of sovereignty, by means of use or occupation, or by any other means." This is the provision the Applicant cited at Opposition footnote 30 for the proposition that space operations are extraterritorial.

Article VI: "States Parties to the Treaty shall bear international responsibility for national activities in outer space, including the Moon and other celestial bodies, whether such activities are carried on by governmental agencies or by non-governmental entities, and for assuring that national activities are carried out in conformity with the provisions set forth in the present Treaty. The activities of non-governmental entities in outer space, including the Moon and other celestial bodies, shall require authorization and continuing supervision by the appropriate State Party to

the Treaty." This is the provision the Applicant did not cite. It is the treaty basis for the FCC's Part 25 licensing authority over commercial satellite systems.

Article VII: "Each State Party to the Treaty that launches or procures the launching of an object into outer space, including the Moon and other celestial bodies, and each State Party from whose territory or facility an object is launched, is internationally liable for damage to another State Party to the Treaty or to its natural or juridical persons by such object or its component parts on the Earth, in air space or in outer space, including the Moon and other celestial bodies." The United States bears international liability for damage caused by the Applicant's satellites, including damage caused by reentry debris. That liability is not extraterritorial. It attaches to the launching state.

Article IX: "States Parties to the Treaty shall pursue studies of outer space, including the Moon and other celestial bodies, and conduct exploration and use of outer space, including the Moon and other celestial bodies, so as to avoid their harmful contamination and also adverse changes in the environment of the Earth resulting from the introduction of extraterrestrial matter and, where necessary, shall adopt appropriate measures for this purpose." The deorbiting of satellites introduces metallic aerosols and other combustion byproducts into the stratosphere. That constitutes the introduction of extraterrestrial matter into the Earth's environment within the meaning of Article IX. The Applicant's own phased deployment admission acknowledges that the atmospheric effects of this process at scale are not yet known. Article IX imposes an affirmative obligation to avoid those effects, not merely to study them after deployment.

C. The Convention on International Liability for Damage Caused by Space Objects (1972)

Convention on International Liability for Damage Caused by Space Objects, Mar. 29, 1972, 24 U.S.T. 2389, 961 U.N.T.S. 187 (Liability Convention).

Article II: "A launching State shall be absolutely liable to pay compensation for damage caused by its space object on the surface of the Earth or to aircraft in flight." This is strict liability. It does not require fault. Debris from deorbiting Orbital Data Center satellites that causes damage on the surface of the Earth or to aircraft triggers absolute liability of the United States as the launching state. The Starship Flight 7 and Flight 8 debris events in the Turks and Caicos and the Bahamas demonstrate that this liability is not theoretical.

Article VII: Claims for compensation may be presented by a state whose natural or juridical persons suffer damage. The United Kingdom has already contacted the United States regarding the Starship trajectory's impact on British overseas territories. See Appendix B, Section B.6.

D. The Registration Convention (1975)

Convention on Registration of Objects Launched into Outer Space, Jan. 14, 1975, 28 U.S.T. 695, 1023 U.N.T.S. 15 (Registration Convention).

Article II: Requires the launching state to register space objects in a national registry and furnish information to the Secretary-General of the United Nations. Article IV specifies the information required, including orbital parameters, general function, and nodal period. One million satellites would constitute the single largest registration obligation in the history of the Convention. The record does not address how this obligation would be discharged or whether the current registration infrastructure can accommodate it.

Record Basis:

The Bakken IO and the Bakken EP each identified the absence of ITU coordination data. Multiple parties independently raised overlapping coordination concerns. The Opposition does not supply this data and does not address its absence.

The Applicant's Opposition at footnote 30 cites the Outer Space Treaty for the proposition that space is extraterritorial. It cited Article II. It did not cite Article VI, VII, or IX. The FCC's Part 25 licensing authority is itself an instrument of Article VI compliance. The Applicant cannot invoke Article II to argue that space operations are beyond the reach of U.S. environmental law while relying on the licensing framework that exists because Article VI requires U.S. authorization and continuing supervision of those operations. The two obligations arise from the same treaty. They cannot be selectively invoked.

Article IX's requirement to avoid adverse changes to the Earth's environment from extraterrestrial matter is directly relevant to the atmospheric effects of industrial-scale satellite reentry. The Applicant admits those effects are not yet quantified. The Treaty imposes an obligation to avoid them, not merely to monitor them after deployment begins.

The Liability Convention's absolute liability regime means the United States bears financial responsibility for damage caused by every satellite in the proposed constellation, on the surface of the Earth and to aircraft in flight, without regard to fault. The record does not contain any assessment of the aggregate liability exposure a million-satellite constellation creates for the United States, any analysis of whether the Applicant's insurance or indemnification is adequate to that exposure, or any indication that the State Department has been consulted regarding the treaty implications of the proposed system.

The Registration Convention requires the United States to register every space object launched from its territory. The record does not address how the registration of up to one million satellites would be accomplished, whether the existing UN Registry infrastructure can accommodate that volume, or what timeline registration would follow.

The ITU coordination obligation, the Outer Space Treaty's authorization and supervision requirement, the Liability Convention's absolute liability regime, and the Registration Convention's registration obligation are each independently relevant to the Commission's public interest determination. The record does not contain sufficient information to evaluate compliance with any of them at the proposed scale.

Petitioner preserves all arguments arising from Executive Order 12114, the ITU Radio Regulations, the Outer Space Treaty (Articles II, VI, VII, and IX), the Liability Convention, and the Registration Convention to the fullest extent allowable by law, including their application to the Commission's environmental review obligations, the scope of its continuing supervision duty, the aggregate liability exposure of the United States, and the public interest determination under 47 U.S.C. Section 309(a).

XII. SECRET LAW PROHIBITION AND ACCEPTANCE STANDARDS

Argument Preserved: To the extent the Commission applied internal, unpublished criteria in its acceptance determination, those criteria constitute agency policy operationally applied to a specific case and must be publicly available under the APA and FOIA.

Supporting Authorities: *NLRB v. Sears, Roebuck & Co.*, 421 U.S. 132 (1975); 5 U.S.C. Section 552(a)(2).

Record Basis: The Application was accepted within approximately two business days during a government shutdown. The ICFS User Guide does not describe internal staff acceptance review standards. See Bakken IR, Request 9.

XIII. STRUCTURAL CONSTITUTIONAL PRESERVATION

Argument Preserved: To the fullest extent allowable by law, Petitioner preserves all structural constitutional objections, including those arising under the Appointments Clause, the Non-Delegation Doctrine, and the Major Questions Doctrine, to the extent that any Commission action in this proceeding is taken by officials whose authority to act is constitutionally deficient or exceeds the scope of congressional delegation. This preservation is precautionary and does not constitute a present assertion that any specific constitutional violation has occurred.

XIV. NOTICE, ACCESS, AND THE COMMISSION'S FILING INFRASTRUCTURE

Argument Preserved: The Commission's filing infrastructure presents structural barriers to meaningful participation by unrepresented parties that bear on the Commission's obligation to provide adequate notice and a reasonable opportunity to be heard.

Supporting Authorities:

The DOTGOV Act of 2020, Division U, Title IX of P.L. 116-260, requires executive-branch agencies to use .gov or .mil domains for official services. The Commission's International Communications Filing System (ICFS) operates on ServiceNow commercial infrastructure at fccprod.servicenowservices.com, not on a .gov domain. The system that governs acceptance, public notice, pleading, and service for the largest satellite application in Commission history does not operate on government infrastructure as Congress directed.

47 C.F.R. Section 25.151 requires public notice sufficient to inform interested parties of pending applications and the deadlines for comment. The adequacy of that notice depends on the accessibility of the system through which participation is accomplished. The ICFS User Guide (June 2025, prepared by ICF, a government contractor) does not describe the internal acceptance review standards, does not explain how validation errors are evaluated, and does not provide guidance on the procedural vehicles available to unrepresented participants for different types of filings. The Commission's own Part 25 Space Station Licensing Process and Timeline webpage provides general guidance but does not address the interaction between ICFS filing categories and the substantive procedural vehicles codified in Sections 25.154, 1.41, 1.45, and 309(d).

5 U.S.C. Section 552(a)(1) requires publication in the Federal Register of descriptions of agency organization, rules of procedure, and substantive rules of general applicability. To the extent the Commission's acceptance criteria, validation standards, or procedural classification rules for ICFS filings are not published, they constitute the kind of unpublished policy the APA prohibits agencies from relying upon adversely. See *NLRB v. Sears, Roebuck and Co.*, 421 U.S. 132 (1975).

Mathews v. Eldridge, 424 U.S. 319 (1976): Due process requires notice reasonably calculated to inform affected parties and an opportunity to be heard that is meaningful in light of the circumstances. A filing system that operates on non-government commercial infrastructure, lacks published procedural guidance for the vehicles it offers, and processed the largest satellite application in Commission history during a partial government shutdown raises questions about whether the notice and participation framework satisfied that standard for unrepresented parties.

Record Basis: Petitioner filed five instruments across three different FCC systems (ICFS, ECFS, and email service) over twelve days. The ICFS system returned validation errors that were not explained. The relationship between ICFS filing categories and the substantive procedural vehicles under Part 25 and Part 1 was not documented in any publicly available guidance Petitioner could locate. The Application itself was accepted within approximately two business days during a government shutdown, on infrastructure operated by a government contractor on a non-.gov domain. See Bakken IR, Request 9. To the extent any of Petitioner's filings were classified, routed, or evaluated under criteria not publicly available, that classification is preserved as a basis for challenge. Petitioner does not assert that a due process violation has occurred. Petitioner preserves the argument that the structural conditions under which this

proceeding was conducted warrant scrutiny, particularly if any filing by any party is excluded or diminished on procedural grounds the filing infrastructure did not make visible.

APPENDIX B: THE FAA RELIANCE GAP

The Applicant's Consolidated Opposition asserts that environmental impacts associated with launch and reentry are already regulated by other agencies, principally the Federal Aviation Administration. This is the core of the Applicant's extraterritorial defense to NEPA: that the FCC need not conduct environmental review because the FAA already has. If the Commission intends to rely on that assertion, the Commission must ask: where is the FAA data on this record?

B.1 What the FAA Has Actually Approved

The FAA's environmental review of Starship operations is a matter of public record. As of the date of this filing, the FAA has approved the following launch cadence for the Starship/Super Heavy vehicle:

- At Boca Chica, Texas (Starbase): up to 25 Starship/Super Heavy launches per year and up to 25 landings per stage per year, pursuant to a Mitigated Finding of No Significant Impact and Record of Decision issued in May 2025.
- At Kennedy Space Center, LC-39A: up to 44 Starship/Super Heavy launches per year and up to 88 landings (44 per stage), pursuant to the Final Environmental Impact Statement and Record of Decision issued January 30, 2026.
- At Cape Canaveral Space Force Station, SLC-37: a separate Environmental Impact Statement proposes up to 76 launches and landings per stage annually. The status of that review is not confirmed on this record.

The combined approved or proposed Starship launch cadence from these three sites is therefore approximately 145 launches per year, assuming all approvals are completed and operational infrastructure is built.

B.2 What the FAA Has Not Approved

The FAA's Environmental Assessments and Environmental Impact Statements evaluate the environmental effects of Starship launches and landings at specific sites, at specific cadences. They do not evaluate, and do not purport to evaluate, the following:

The cumulative environmental effect of launching, deploying, maintaining, and replacing a fleet of up to one million Orbital Data Center satellites over the system's operational lifetime. The lifecycle throughput of a million-satellite constellation is not bounded anywhere in the FAA's environmental reviews or anywhere on this Commission's record.

The atmospheric effects of reentry at the scale described. The FAA's environmental reviews do not evaluate the cumulative stratospheric loading from the deorbiting and atmospheric ablation of hundreds of thousands of satellites per year at steady state.

The environmental effects of the Orbital Data Center system as a system. The FAA licenses launch vehicles and launch sites. It does not license satellite constellations. That evaluation, to the extent any agency conducts it, falls to the FCC under 47 C.F.R. Section 1.1307.

B.3 The Arithmetic the Record Does Not Contain

The Commission should have before it the answers to straightforward questions the record does not address:

How many Orbital Data Center satellites does each Starship launch deliver? The record does not state this.

How many launches per year are required for initial deployment? For steady-state replacement? For end-of-life disposal operations? The record does not contain these figures.

What fraction of the FAA-approved Starship launch cadence would the Orbital Data Center system consume? If the answer is a substantial fraction, the system's deployment competes with every other mission Starship is intended to serve.

B.4 Where FAA Jurisdiction Ends

The FAA's jurisdiction under the Commercial Space Launch Act, 51 U.S.C. Sections 50901-50923, extends to commercial launch and reentry activities. It does not cover what happens between launch and reentry. It does not cover orbital operations. It does not cover the cumulative effects of a constellation of one million satellites operating in low Earth orbit for decades. It does not cover the aggregate atmospheric chemistry effects of deorbiting those satellites at end of life.

The space between launch and reentry is precisely where the Orbital Data Center system exists. It is also precisely where the FAA's jurisdiction does not reach. If the Commission accepts the extraterritorial argument and simultaneously declines to conduct its own environmental review, the result is a regulatory gap in which no federal agency evaluates the system-level environmental effects of a million-satellite constellation. That gap is the structural consequence of the Applicant's own legal position.

B.5 What the Commission Must Require

If the Commission intends to rely on the FAA's environmental process, the Commission should require, at minimum, that the following information be placed on the record: the specific FAA launch licenses, EAs, EISs, and RODs the Applicant contends cover the launch and reentry activities; the projected launch cadence for initial deployment, steady-state maintenance, and

end-of-life operations; the current status of the Starship vehicle operator license; and a clear delineation of where FAA jurisdiction reaches and where it ends with respect to the proposed system.

Without this information, the Commission is being asked to rely on another agency's environmental process without knowing what that process actually evaluated. A decision grounded in an assumption that another agency has already done the work, without verifying what that agency actually did, is not a decision grounded in the record. It is a decision grounded in blind faith.

B.6 The Demonstrated Flight Record

The Applicant's Consolidated Opposition invokes its operational track record as a basis for institutional trust. The Commission should evaluate that invocation against the demonstrated Starship flight record as of the date of this filing.

Flight 7 (January 16, 2025): Upper stage destroyed during ascent. Debris landed on the Turks and Caicos Islands, including one confirmed instance of property damage (a vehicle struck on South Caicos). FAA mishap investigation opened.

Flight 8 (March 6, 2025): Upper stage destroyed during ascent. Debris landed near the Bahamas and the Turks and Caicos Islands. FAA mishap investigation opened. The United Kingdom government minister responsible for overseas territories contacted the United States regarding the trajectory's impact on British territories.

Flight 9 (May 27, 2025): FAA issued return-to-flight determination on May 22, 2025, while the Flight 8 mishap investigation remained open. Upper stage completed its burn and reached its intended trajectory but failed during the coast phase, losing attitude control. Vehicle destroyed. Third consecutive upper stage failure.

Flight 10 (August 26, 2025): First successful mission after three consecutive upper stage failures. Deployed eight Starlink simulators. Booster splashed down in Gulf of Mexico. Ship completed controlled splashdown in Indian Ocean.

Flight 11 (October 13, 2025): Successful mission. Last flight of 2025. Reused a booster previously flown on Flight 8.

No additional Starship flights occurred in the remainder of 2025. Starship Version 3 is in preparation for its first flight as of March 2026.

The FAA approved 25 Starship launches per year from Boca Chica in May 2025. Five Starship flights actually occurred in 2025. Three of those five resulted in upper stage destruction. The demonstrated cadence is 20% of the approved cadence. The demonstrated success rate for the upper stage in 2025 is 40%. The record before this Commission does not contain any analysis reconciling the proposed system's deployment requirements with this demonstrated launch performance, nor any projection of when the launch cadence necessary to deploy, maintain, and replace a million-satellite constellation will be achievable.

B.7 The Commission's Independent Environmental Obligation: Section 1.1307(c), Executive Order 12114, and the Outer Space Treaty.

B.7 The Commission's Independent Environmental Obligation: Section 1.1307(c), Executive Order 12114, the Outer Space Treaty, and the Procedural Framework the Opposition Did Not Address.

The Applicant's environmental defense addresses NEPA. The Bakken EP was filed under 47 C.F.R. Section 1.1307(c). These are not the same instrument, they do not share the same legal basis, and the Applicant's arguments against one do not reach the others.

B.7.A. The Commission's Own Rule Is Not NEPA.

Section 1.1307(c) is the Commission's own regulation. It provides that if an interested person alleges a categorically excluded action will have a significant environmental effect, the Bureau "shall review the petition and consider the environmental concerns that have been raised." The rule uses "shall." The Bureau's obligation is mandatory, not discretionary. The rule does not condition that obligation on NEPA being the source of the environmental concern. It requires the Bureau to consider "the reasons justifying or circumstances necessitating environmental consideration in the decision-making process." Environmental consideration may be necessitated by sources independent of NEPA: the Commission's public interest obligation under 47 U.S.C. Section 309(a), by Executive Order 12114, by Article IX of the Outer Space Treaty the Applicant itself invoked, by the Applicant's own phased deployment admission that atmospheric effects are unknown, or by the peer-reviewed science on the record.

Section 1.1307(c) remains in effect until amended or repealed through notice-and-comment rulemaking. The Commission's August 2025 NEPA modernization NPRM proposes to exclude satellite operations from NEPA review. It has not done so. Even if it does, the NPRM addresses NEPA. It does not propose to repeal Section 1.1307(c) as applied to non-NEPA environmental obligations.

The Opposition does not argue that Section 1.1307(c) is inapplicable. It does not argue that the Bureau has no obligation to review the Bakken EP. It argues that NEPA does not apply. That argument, even if correct, does not reach the rule under which the Bakken EP was filed.

Even absent the Bakken EP, the Bureau retains independent authority under Section 1.1307(d) to require an EA on its own motion if it determines that a categorically excluded action may have significant environmental impact. The Applicant's own phased deployment admission, the peer-reviewed science on the record, NASA's concerns, as set forth in its letter filed in this docket and analyzed at Appendix H, Section H.3, independently provide the basis for the Bureau to act under this provision.

B.7.B. The "Prior to Authorization" Timing Obligation.

Section 1.1313(b) provides: "Informal objections which are based on environmental considerations must be filed electronically prior to grant of the construction permit, or prior to authorization for facilities that do not require construction permits."

Space stations do not require construction permits. 47 C.F.R. Section 25.113(f). The ODC Application is therefore subject to the "prior to authorization" clause. The Bakken EP was filed March 9, 2026. The Application has not been authorized. The petition is timely. The Bureau's obligation to review and consider the environmental concerns raised in the Bakken EP, and to determine whether the action may have a significant environmental impact, must be discharged before the Commission grants the authorization. The Opposition asks the Commission to "reject calls for a lengthy environmental review." Section 1.1313(b) does not permit the Commission to grant the authorization while a timely environmental objection remains unresolved.

The Bakken EP was filed under both Section 1.1313(a), as a petition to deny concurrently filed under 47 U.S.C. Section 309(d), and Section 1.1313(b), as an informal objection based on environmental considerations. Either procedural vehicle requires resolution before grant.

B.7.C. The FAA Reliance Clause Does Not Apply.

The Opposition relies on 47 C.F.R. Section 1.1311(e), which provides that an EA need not be submitted to the Commission if another agency "has assumed responsibility for determining whether the facilities in question will have a significant effect on the quality of the human environment."

The operative phrase is "the facilities in question." The facilities in question are one million Orbital Data Center satellites operating in low Earth orbit for decades. The FAA has not assumed responsibility for determining the environmental effects of those facilities. The FAA has evaluated Starship launch cadences from specific sites at specific rates. It has not evaluated the ODC constellation as a system. It has not evaluated atmospheric loading from approximately 200,000 satellite reentries per year at steady state. It has not evaluated the cumulative lifecycle throughput of a million-satellite constellation. No federal agency has. Section 1.1311(e) does not relieve the Commission of its obligation when no other agency has assumed the responsibility the rule describes, for the specific facilities this Application proposes.

B.7.D. The Applicant's Own Admission Anticipates the Discovery Clause.

Section 1.1312(d) provides: "If, following the initiation of construction under this section, the licensee or applicant discovers that the proposed facility may have a significant environmental effect, it shall immediately cease construction which may have that effect, and submit the information required by Section 1.1311."

The Applicant's phased deployment admission at Opposition page 16 states that it will "monitor actual atmospheric effects, validate models with real-world data, and implement any necessary adjustments as it scales." That statement is a pre-authorization acknowledgment that the Applicant expects to discover environmental effects during deployment that it cannot presently quantify. Section 1.1312(d) would require the Applicant to cease deployment and submit an EA at the point of discovery. The Commission should consider whether it is appropriate to authorize a system the Applicant itself predicts will trigger the discovery clause, rather than requiring the environmental evaluation the Commission's own rules contemplate before authorization.

B.7.E. The Applicant's Own Concession Triggers Executive Order 12114.

The Opposition argues that the 2023 Fiscal Responsibility Act excludes "extraterritorial activities or decisions with effects located entirely outside of the jurisdiction of the United States" from NEPA's definition of "major federal action." Opposition at 17. In making this argument, the Applicant concedes two facts: first, that the FCC's licensing of a million-satellite constellation is a major federal action significant enough to engage the statutory framework; and second, that the effects of that action are environmentally significant and occur, in the Applicant's own framing, outside the jurisdiction of the United States.

The Applicant's defense to NEPA is that the effects are extraterritorial. That defense does not reach Executive Order 12114. It triggers it.

Executive Order 12114, signed January 4, 1979, operates independently of NEPA under the President's constitutional authority over foreign policy. Section 2-3(a) requires federal agencies to evaluate the environmental effects of major federal actions "significantly affecting the environment of the global commons outside the jurisdiction of any nation." NASA's formal implementing guidance defines global commons to include "the oceans and the atmosphere." The upper atmosphere and stratosphere, where satellite reentry deposits metallic aerosols at the scale the Applicant's own lifecycle throughput implies, are global commons by any definition.

EO 12114 is not NEPA. NASA's own guidance confirms: "EO 12114 is not NEPA or the implementation of NEPA; however, EO 12114 does further the purpose of NEPA." The Fiscal Responsibility Act of 2023 amended NEPA's statutory definition of major federal action. It did not amend, repeal, or supersede EO 12114, which derives from independent executive authority.

The Opposition cites Executive Order 14154 (Unleashing American Energy, January 20, 2025) at footnote 31 as further support for NEPA's inapplicability. EO 14154 contains a specific revocation list. It revokes EO 11991 (Carter's 1977 order empowering CEQ) and thirteen Biden-era executive orders, each identified by number. EO 12114 is not among them. It was not revoked. It remains in effect. The Department of Defense maintains active implementing regulations at 32 C.F.R. Part 187. NASA complies with EO 12114 for its own space missions, including the Mars Sample Return Campaign, for which NASA prepared a Programmatic Environmental Impact Statement with an EO 12114 compliance checklist. The FCC licenses the same category of space activity that NASA evaluates under EO 12114. The record contains no indication that the Commission has evaluated its EO 12114 obligations with respect to this Application.

If the effects are domestic, NEPA applies. If the effects are extraterritorial on the global commons, EO 12114 applies. There is no third option in which neither applies. The Applicant's defense concedes the predicates of the authority it did not address.

B.7.F. The Outer Space Treaty Independently Requires Environmental Consideration.

The Opposition cites the Outer Space Treaty at footnote 30 for the proposition that space is extraterritorial. It cites Article II. It does not cite Article IX, which provides: "States Parties to the Treaty shall pursue studies of outer space... so as to avoid their harmful contamination and also adverse changes in the environment of the Earth resulting from the introduction of extraterrestrial matter and, where necessary, shall adopt appropriate measures for this purpose."

Satellite reentry introduces metallic aerosols and combustion byproducts into the stratosphere. That constitutes the introduction of extraterrestrial matter into the Earth's environment within the meaning of Article IX. The Applicant's own phased deployment admission acknowledges the atmospheric effects at scale are not yet known. Article IX imposes an affirmative obligation to avoid those effects, not merely to monitor them after deployment begins. The Applicant invoked the Treaty. It cannot selectively cite Article II while ignoring Article IX of the same instrument.

B.7.G. Convergence: Three Independent Sources, One Mandatory Obligation.

The Commission's environmental obligation in this proceeding arises from three independent sources, each sufficient standing alone: (1) Section 1.1307(c), the Commission's own rule, which mandates Bureau review of the Bakken EP before authorization under Section 1.1313(b) and is not conditioned on NEPA; (2) Executive Order 12114, which independently requires environmental consideration of major federal actions affecting the global commons, which the Applicant's own extraterritorial argument concedes; and (3) Article IX of the Outer Space Treaty, which the Applicant itself invoked while ignoring its environmental provisions. The Commission's own procedural rules reinforce this obligation: Section 1.1311(e) does not relieve the Commission because no other agency has assumed responsibility for these facilities; Section 1.1312(d) will be triggered by the Applicant's own predicted discovery of environmental effects; and Section 1.1313(b) prohibits authorization while the Bakken EP remains unresolved.

To the extent the Commission determines that NEPA does not require environmental review of this Application, the Commission should state on the record whether Section 1.1307(c), Executive Order 12114, and Article IX of the Outer Space Treaty independently require it to consider the atmospheric effects of approximately 200,000 satellite reentries per year before grant. The Commission should also state whether it has evaluated the applicability of Section 1.1311(e) to the specific facilities this Application proposes and whether the FAA has in fact assumed the responsibility that provision describes. To the extent any Commission rule, categorical exclusion, or pending rulemaking operates to prevent the preparation of an Environmental Assessment for this Application, Petitioner requests waiver of such rule under 47 C.F.R. Section 1.3, with good cause shown by the Executive Order 12114 and Outer Space Treaty obligations bearing on this Application, which the Commission cannot displace by regulation and which independently require environmental consideration prior to authorization. Petitioner preserves all arguments arising from Sections 1.1307(c), 1.1307(d), 1.1311(e), 1.1312(d), 1.1313(a) and (b), Executive Order 12114, and Articles VI and IX of the Outer Space Treaty for reconsideration under 47 U.S.C. Section 405 and judicial review under 47 U.S.C. Section 402(b) and 5 U.S.C. Section 706.

APPENDIX C: THE STARSHIELD SPECTRUM RECORD

C.1 The Documented Transmissions.

In October 2025, L. Scott Tilley, an amateur satellite tracker and physicist based in British Columbia, Canada, detected persistent space-to-Earth transmissions from approximately 170

satellites in SpaceX's Starshield constellation operating in the 2025 to 2110 MHz frequency band. Tilley published his findings on the open-access research repository Zenodo, documenting that the emissions were "centred within the 2025-2110 MHz band allocated internationally for Earth-to-space and space-to-space links" and exhibited "Doppler characteristics and power levels consistent with direct space-to-Earth transmissions." L. Scott Tilley, "Detection of Strong S-Band Emissions from the Starshield Constellation: Observations and Regulatory Context," Zenodo (Oct. 17, 2025), DOI: 10.5281/zenodo.17373141. The signals spanned tens of megahertz in bandwidth, were coherent and Doppler-clean, and remained detectable in antenna side-lobes, properties the author found inconsistent with deep-space or cross-link activity. *Id.*

The 2025-2110 MHz band is reserved under ITU Radio Regulations for Earth-to-space uplink transmissions. ITU Radio Regulation Footnote 5.392 reserves this spectrum primarily for Earth-to-space and space-to-space links. ITU-R Recommendation SA.1154 specifies that the band is not intended for space-to-Earth data transfer due to the risk of radio-frequency interference with co-located satellites. Downlinking on an uplink-reserved band contravenes the allocation framework designed to prevent interference with other satellites' ability to receive commands from ground stations.

The findings were reported by NPR on October 17, 2025, and subsequently confirmed by multiple independent sources. Geoff Brumfiel, "A classified network of SpaceX satellites is emitting a mysterious signal," NPR (Oct. 17, 2025). Kevin Gifford, a computer science professor at the University of Colorado Boulder specializing in spacecraft radio interference, stated to NPR that the interference was "definitely happening," though the extent of disruption remained unresolved. *Id.* Tilley warned that "[n]earby satellites could receive radio-frequency interference and could perhaps not respond properly to commands, or ignore commands, from Earth." *Id.* Neither SpaceX nor the National Reconnaissance Office responded to NPR's request for comment. *Id.*

Starshield is a classified satellite constellation built by SpaceX and operated by the U.S. National Reconnaissance Office under a \$1.8 billion contract first reported by Reuters in 2024. The NRO has conducted at least eleven launches of Starshield satellites since May 2024, building what the NRO describes as a "proliferated system" of over 200 satellites designed for rapid Earth observation and resilient communications. The Starshield network shares design heritage with SpaceX's commercial Starlink constellation but adds classified cryptographic capability.

As of the date of this filing, no publicly visible enforcement action, investigation, or formal inquiry has been opened by the FCC, NTIA, or any other agency regarding the Starshield transmissions.

C.2 Enforcement Referral Pathways.

The regulatory landscape for addressing the Starshield transmissions is divided across three jurisdictional bodies. To the extent the transmissions involve federal government spectrum use coordinated through NTIA, NTIA is the appropriate enforcement body under 47 U.S.C. Sections 901-904. To the extent the transmissions affect non-federal spectrum users or violate international spectrum allocations under the ITU Radio Regulations, the Commission's Enforcement Bureau has authority under 47 U.S.C. Sections 312 and 503(b). To the extent the transmissions involve classified national security programs, the Department of Defense Inspector General has oversight authority. The Commission should determine which enforcement pathway is appropriate based on the information available to it, including any classified information not publicly visible, and should state its determination on the record.

Petitioner preserves all arguments arising from the Starshield frequency record to the fullest extent allowable by law, including their application to the Commission's character and candor evaluation, the weight assigned to the Applicant's unsworn spectrum compliance representations, the enforcement referral analysis, and the public interest determination under 47 U.S.C. Section 309(a). To the extent the Commission determines that the Starshield matter has been resolved through channels not publicly visible, Petitioner respectfully requests that the Commission state that determination on the record so that all parties may evaluate it.

C.3 The Trilemma the Record Presents.

The Starshield frequency record presents the Commission with three possible explanations. Each carries national security implications that require investigation regardless of which is correct.

Explanation One: Authorized classified operations in violation of the allocation framework.

If the transmissions are authorized government operations conducted under classified authorities and coordinated through NTIA, those operations must still comply with the spectrum allocation framework. Executive Order 13526, Section 1.7(a)(1), provides that "[i]n no case shall information be classified, continue to be maintained as classified, or fail to be declassified in order to: (1) conceal violations of law, inefficiency, or administrative error." 75 Fed. Reg. 707 (Jan. 5, 2010). Classified status does not exempt government operations from the ITU Radio Regulations or the domestic spectrum allocation table. Federal government spectrum use is coordinated through NTIA under 47 U.S.C. Sections 901-904 precisely to prevent interference with other users.

Downlinking on a band reserved for Earth-to-space uplink contravenes the allocation framework regardless of the classification level of the program. ITU-R Recommendation SA.1154 establishes that the 2025-2110 MHz band is not intended for space-to-Earth transmissions specifically because such use risks radio-frequency interference with co-located satellites'

command reception. A classified program that violates spectrum allocations requires more scrutiny, not less, because the interference it causes to other operators is indistinguishable from unauthorized commercial use and the affected parties have no mechanism to identify or resolve the source.

If this is the explanation, the Commission should determine whether NTIA authorized the specific frequency usage documented by Tilley and whether that authorization complies with the applicable spectrum allocation framework.

Explanation Two: Unauthorized transmissions from a national security asset.

If the transmissions are not authorized under any government spectrum coordination, they constitute unlicensed transmissions from a national security asset operating outside the framework the Commission and NTIA jointly administer under 47 U.S.C. Sections 901-904 and the NTIA Manual of Regulations and Procedures for Federal Radio Frequency Management. An entity operating a classified constellation without proper spectrum coordination presents a national security concern: it means a satellite system with access to classified government networks is transmitting on frequencies it is not authorized to use, in a manner that neither the Commission nor NTIA appears to have detected or addressed on any publicly visible record.

If this is the explanation, the implications extend beyond spectrum enforcement. They implicate the integrity of the government's own oversight of the classified systems it has entrusted to this Applicant. The NRO's stated mission for Starshield is to deliver data "in minutes or even seconds" through a resilient, distributed architecture. If that architecture is operating outside the spectrum coordination framework, the framework itself is compromised.

Explanation Three: Compromise or impersonation of a classified constellation's emissions signature.

If the detected transmissions do not originate from the Starshield constellation at all, then the emissions signature of a classified national security constellation providing communications services to the United States government has been spoofed, impersonated, or falsely attributed by an unknown source. Tilley's analysis documented that the signals exhibited orbital characteristics consistent with approximately 70-degree inclination, approximately 14.97 revolutions per day orbits, matching known Starshield orbital parameters. Zenodo, *supra*. The signals were coherent, Doppler-clean, and detectable in antenna side-lobes, properties the researcher found consistent with direct satellite-to-Earth transmissions and inconsistent with deep-space or cross-link activity. *Id.*

If a third party or unknown source is capable of producing signals with these characteristics in a manner that is publicly reported as authentic and attributed to a classified constellation, without detection or correction by the operator, the NRO, NTIA, or the Commission, that represents a

vulnerability in the classified network the responsible agencies must evaluate. The alternative explanation, that the attribution is in error, does not reduce the national security concern. It heightens it: either the classified constellation is transmitting where it should not be, or someone is making it appear that it is, and neither the operator nor the oversight agencies have publicly addressed either possibility.

Neither SpaceX nor the NRO responded to media inquiries. NPR (Oct. 17, 2025). If the transmissions are not what they appear to be, the silence is more concerning, not less.

C.4 The Convergence Across All Three Explanations.

Each explanation requires investigation. Under the first, the spectrum framework has been violated under the cover of classification, in tension with Executive Order 13526, Section 1.7(a)(1). Under the second, a national security asset is operating outside the regulatory framework the government created to manage exactly this risk. Under the third, a classified constellation's emissions signature has been compromised or impersonated without detection or response by any responsible agency.

None of these is trivial. None is political. Each is a national security matter requiring coordination between the Commission, NTIA, and the Department of Defense before the Commission authorizes one million additional satellites sharing design heritage and infrastructure with the constellation in question. The ODC Application proposes to deploy a system more than one hundred times the size of Starshield, using the same corporate infrastructure, the same launch vehicles, and the same design lineage. The spectrum compliance record of the smaller constellation bears directly on the reliability of representations made for the larger one.

The Opposition was silent. The record contains no acknowledgment, no explanation, and no resolution. Silence is not resolution, particularly where the Applicant demands enforcement against competitors for their spectrum conduct while its own constellation's spectrum conduct remains unexplained on any publicly visible record. The Commission has the authority and the obligation to determine which explanation applies. 47 U.S.C. Sections 154(i), 303, 312. To the extent the answer involves classified information not appropriate for the public record, the Commission should coordinate with NTIA and the Department of Defense through appropriate channels and state on the public record that the coordination occurred and the matter was resolved.

C.5 Preservation.

Petitioner preserves all arguments arising from the Starshield frequency record to the fullest extent allowable by law, including but not limited to their application to: the Commission's character and candor evaluation under 102 F.C.C.2d 1179; the weight assigned to the Applicant's

unsworn spectrum compliance representations in the Opposition; the enforcement referral analysis under 47 U.S.C. Sections 312 and 503(b); the hearing designation analysis under 47 U.S.C. Section 309(e); and the public interest determination under 47 U.S.C. Section 309(a). To the extent the Commission resolves this matter through classified channels, Petitioner requests that the Commission state on the public record that the coordination occurred, identify the agencies involved, and confirm that the matter was resolved to the Commission's satisfaction before final action on this Application.

APPENDIX D: THE ITAR COMPLIANCE QUESTION AND THE EPSTEIN FILES

This Appendix presents a narrow compliance question, not a narrative. The question is whether the threshold for an ITAR compliance evaluation has been reached on the facts now in the federal record. Under the ITAR framework, that threshold is not high. The presence of any foreign person at a facility where USML-controlled articles are manufactured or accessible triggers a compliance evaluation obligation. 22 C.F.R. Sections 120.16, 120.54, 127.12. The failure to conduct or disclose that evaluation is not a minor regulatory omission. ITAR violations carry civil penalties of up to \$500,000 per violation under 22 U.S.C. Section 2778(e) and criminal penalties of up to \$1,000,000 and 20 years imprisonment for willful violations under 22 U.S.C. Section 2778(c). The State Department's Directorate of Defense Trade Controls has imposed consent agreements exceeding \$100 million in recent enforcement actions against defense contractors for unauthorized transfers and disclosure failures. The facts documented below are drawn entirely from federal records released by the United States Department of Justice and from independent reporting confirmed by multiple outlets. No characterization of any person's private conduct is offered. The relevance to this proceeding is limited to the Commission's character and compliance evaluation under 47 U.S.C. Section 308 and 102 F.C.C.2d 1179.

D.1 What the Federal Record Now Establishes

On January 30, 2026, the United States Department of Justice began releasing documents pursuant to the Epstein Files Transparency Act of 2025. The released documents include at least sixteen emails between Jeffrey Epstein and Elon Musk exchanged between 2012 and 2014, as well as dozens of additional emails between their respective assistants. These documents are federal records released by the United States government. Their authenticity has not been disputed. Musk acknowledged the emails on X on February 1, 2026, stating that he had "very little correspondence with Epstein" and that the emails "could be misinterpreted."

The SpaceX facility visit was Musk's idea. In February 2013, Musk suggested that Epstein visit the SpaceX factory near Long Beach, California. See *Fortune* (Jan. 30, 2026); *Byline Times* (Feb. 17, 2026). Musk's assistant, Mary Beth Brown, confirmed the luncheon for February 25,

2013 at 1:00 PM. A message alert in the DOJ files confirms the Epstein-Musk lunch at SpaceX took place on that date. See Byline Times (Feb. 17, 2026).

On February 26, 2013, Epstein emailed Musk: "thanks for the tour." He added: "you would have had fun at xmas." Musk replied with a smiley face emoji. See TIME (Jan. 31, 2026); NBC News (Feb. 13, 2026); Fortune (Jan. 30, 2026). The email constitutes a direct, contemporaneous acknowledgment from the visitor that a tour occurred and a direct, contemporaneous response from the host that does not dispute it.

On July 4, 2020, Musk posted on X: "To the best of our knowledge, he never toured SpaceX. Don't know where that comes from." The January 2026 DOJ document release directly contradicts that statement.

D.2 The Foreign Nationals and the ITAR Framework

Epstein attempted to bring three women who were foreign nationals to the facility: two Russian nationals and one South African national. SpaceX's assistant requested passport scans. After review, she stated the foreign nationals could not be cleared for a full tour but were "welcome to come into the lobby area/conference room adjacent to the lobby."

Epstein resisted the restriction. A separate email from Groff states: "JE (Jeffrey Epstein) is being JE. He says we will go to the lobby. I said I would let Mary Beth know, and he said, 'No! Don't say anything' (in an aggravated voice). So I can't get back to her." See Cybernews (Feb. 23, 2026). This email documents an affirmative attempt by the visiting party to circumvent the facility's access restrictions for uncleared foreign nationals at an ITAR-regulated site.

SpaceX's Hawthorne headquarters manufactures rockets and spacecraft classified under USML Categories IV (Launch Vehicles) and XV (Spacecraft and Related Articles). The company's own hiring practices, documented in the DOJ's 2023 complaint under the Immigration and Nationality Act, confirm that SpaceX restricts facility access to U.S. persons specifically because of ITAR.

D.3 The Two Independent ITAR Triggers.

The ITAR compliance question presents on two independent grounds, each sufficient standing alone. Each created a separate obligation to evaluate and, where warranted, disclose under 22 C.F.R. Section 127.12. The record does not reflect that either obligation was discharged.

D.3.1. The Hawthorne Facility Visit (February 2013).

Four foreign nationals were present at SpaceX's Hawthorne headquarters. Under ITAR, the presence of any foreign person at a facility where controlled articles are manufactured or accessible triggers a compliance evaluation obligation, regardless of what that person saw, regardless of whether they were confined to the lobby, and regardless of any other circumstance. 22 C.F.R. Sections 120.16, 120.54.

The compliance perimeter is the building, not the production floor. This is not Petitioner's characterization. It is the Applicant's own sworn representation to a federal court. In *Space Exploration Technologies Corp. v. Garland*, No. 7:23-cv-00261, Complaint at 2 (S.D. Tex., filed Sept. 15, 2023), SpaceX stated: "Every SpaceX employee has access to technology and data controlled by these statutory and regulatory regimes," including ITAR. If every employee at every SpaceX facility has access to controlled technology, then any foreign person at any SpaceX facility is within the compliance perimeter. The company cannot argue otherwise without contradicting its own sworn pleading.

The DOJ's own complaint independently confirmed the same posture. In *United States v. Space Exploration Technologies Corp.*, No. 23-INA-00094, Complaint at 3-5 (OCAHO, filed Aug. 24, 2023), the DOJ alleged that SpaceX "routinely discouraged asylees and refugees from applying and refused to hire or consider them, because of their citizenship status" and that SpaceX recruiters "regularly told job candidates that with a few exceptions, SpaceX is only able to hire U.S. citizens or lawful permanent residents due to ITAR." SpaceX's own hiring rejection codes included "not authorized to work/ITAR ineligible." The company's facility-wide citizenship restrictions confirm that it treats the facility boundary as the ITAR compliance perimeter for purposes of access, hiring, and visitor clearance.

SpaceX manufactures articles classified under USML Category IV (Launch Vehicles, Guided Missiles, Ballistic Missiles, Rockets, Torpedoes, Bombs, and Mines) and Category XV (Spacecraft and Related Articles). 22 C.F.R. Section 121.1, Categories IV and XV.

The lobby of a rocket factory is still inside a rocket factory. The foreign nationals were inside that perimeter. That fact alone requires evaluation and, where warranted, disclosure under 22 C.F.R. Section 127.12. The record contains no indication that any such evaluation or disclosure occurred

D.3.2. The Austrian Passport (July 2019).

Epstein, the American citizen who arranged and conducted the tour was later found, upon execution of a federal search warrant on July 6, 2019, to possess a stamped Austrian passport under the alias "Marius Fortelni." The passport bears Epstein's photograph. It lists Saudi Arabia as his place of residence. It contains entry and exit stamps from France, Spain, the United Kingdom, and Saudi Arabia, including a stamp from the Saudi Arabian consulate in Vienna. Federal prosecutors introduced the passport in open court during Epstein's July 2019 bail hearing. Its existence, its contents, and its use for international travel are undisputed matters of federal judicial record. *See* DOJ document release (December 2025-January 2026); NBC News (July 15, 2019); The Daily Beast (December 23, 2025).

When the passport entered the federal record in July 2019, it established that the individual who toured SpaceX's ITAR-regulated facility in February 2013, at the personal invitation of SpaceX's

controlling principal, carried a foreign identity document listing Saudi Arabian residence. That revelation created an independent ITAR compliance evaluation moment, separate from and in addition to the evaluation triggered by the facility visit itself.

Saudi Arabia is not a proscribed country under ITAR Section 126.1, but transfers of defense articles and technical data to Saudi Arabia require individual export licenses. The presence at an ITAR-regulated facility of an individual who carried a fraudulent passport listing Saudi Arabian residence, who had extensive documented ties to Saudi government officials at the highest levels (*see* Section D.4), who brought foreign nationals of Russian and South African nationality, and who affirmatively attempted to circumvent the facility's access restrictions, presents a compliance evaluation question that ITAR requires the regulated entity to assess, document, and where warranted, disclose.

A responsible ITAR compliance program would have evaluated whether the original 2013 visitor access assessment was adequate in light of the passport's Saudi dimensions and whether a retroactive voluntary disclosure to DDTC was warranted. 22 C.F.R. Section 127.12. The record before this Commission contains no indication that any such evaluation was conducted: not in 2013 when the foreign nationals accessed the facility, not in 2019 when the passport and its Saudi dimensions entered the federal record, and not in 2026 when the DOJ released the contemporaneous emails contradicting the controlling principal's public denial. The one thing the record does have is a public post denying such a visit to SpaceX ever took place. Federal records show that statement is now undeniably false and further unrebutted in this very proceeding.

D.4 The Saudi Arabian Dimension

Epstein's connections to Saudi Arabia are extensively documented in the released files. The Crown Prince of Saudi Arabia, Mohammed bin Salman, is named ninety-seven times in the released documents. A photograph of Epstein with bin Salman was found displayed in Epstein's Manhattan residence. Emails document planned visits to Saudi Arabia, travel to Riyadh, a gift of a Bedouin tent from the Crown Prince, and coordination through the Saudi Consulate referencing an invitation from the Crown Prince by name. See CBS News (Nov. 19, 2025); LA Magazine (March 2026).

Saudi Arabia is not a proscribed country under ITAR Section 126.1, but transfers of defense articles and technical data to Saudi Arabia require individual export licenses. The presence at an ITAR-regulated facility of an individual carrying a fraudulent passport listing Saudi Arabian residence, who had extensive documented ties to Saudi government officials at the highest levels, who brought foreign nationals of Russian and South African nationality, and who affirmatively attempted to circumvent the facility's access restrictions, presents a compliance evaluation question that ITAR requires the regulated entity to assess, document, and where warranted, disclose.

D.5 The Broader Relationship

The DOJ files establish that the relationship between Musk and Epstein extended well beyond a single facility visit. Musk exchanged at least sixteen direct emails with Epstein and more than fifty total communications were documented. See Fortune (Jan. 30, 2026). Byline Times, citing the DOJ files, confirmed at least four separate in-person meetings between Musk and Epstein after Epstein's 2008 conviction. See Byline Times (Feb. 17, 2026).

Petitioner makes no characterization of any person's conduct beyond what the federal records establish. The relevance to this proceeding is narrow and specific: the controlling principal of the Applicant entity had a documented, multi-year relationship with an individual who brought uncleared foreign nationals to an ITAR-regulated facility operated by the Applicant, who carried an Austrian passport listing Saudi Arabian residence, and who had extensive documented ties to foreign government officials. These facts bear on the Commission's evaluation of the Applicant's character, candor, and compliance posture under 47 U.S.C. Section 308 and the Commission's 1986 character policy, 102 F.C.C.2d 1179.

D.6 The Self-Disclosure Timeline

Under the ITAR voluntary disclosure framework at 22 C.F.R. Section 127.12, DDTC strongly encourages immediate disclosure upon discovery of a potential violation. A disclosure qualifies for mitigating treatment only if the information reaches DDTC prior to government knowledge from another source. 22 C.F.R. Section 127.12(b)(2). Failure to disclose is an adverse factor. 22 C.F.R. Section 127.12(a).

The timeline presents two independent disclosure windows. The first opened on February 25, 2013, the date of the facility visit. The second opened on July 15, 2019, when the passport entered the federal record. The DOJ released the documents on January 30, 2026. Bloomberg reported the foreign nationals' presence on February 23, 2026. The Bakken IO placed the ITAR compliance question on this Commission's record on March 6, 2026. See Bakken IO at 19 (Section III, "Fourth"). The Opposition was filed March 16, 2026. No disclosure, acknowledgment, or response appears on any record before any agency, as far as publicly available information reflects.

D.7 Enforcement Referral Pathways for ITAR

Petitioner does not ask the Commission to adjudicate ITAR compliance. That authority rests with DDTC (civil enforcement, 22 C.F.R. Part 128) and DOJ (criminal enforcement, 22 U.S.C. Section 2778). The appropriate referral is to the Directorate of Defense Trade Controls, Office of Defense Trade Controls Compliance (DTCC), United States Department of State. The Commission has independent referral authority under 47 U.S.C. Sections 154(i) and 303.

D.8 What the Commission Must Know

The Commission is not being asked to resolve the Epstein investigation, adjudicate ITAR compliance, or make findings about any person's private conduct. The Commission is being asked to recognize that publicly available federal records, released by the United States Department of Justice during the active pleading window of this proceeding, establish facts bearing directly on the character, candor, and compliance posture of the Applicant and its controlling principal. Those facts were placed on this Commission's record by the Bakken IO on March 6, 2026. The Applicant's Consolidated Opposition, filed ten days later, did not address them. The Commission need not resolve the ITAR question to recognize that the threshold for evaluation has been reached. It must determine whether that threshold, combined with the Starshield frequency record and the Section 1.65 ownership violation, warrants referral, supplementation, or a condition requiring sworn responses before the Commission acts on this Application.

D.9 The Procedural Relevance to This Proceeding.

This Appendix does not present an enforcement theory. It presents a threshold compliance question that intersects directly with the Commission's statutory obligations under 47 U.S.C. Sections 308 and 309.

The Commission's public interest determination necessarily assumes that the applicant operates in compliance with applicable federal regulatory regimes, particularly those governing foreign access to controlled technologies. Where the record presents facts that independently trigger evaluation obligations under such regimes, the absence of any indication that those obligations were discharged creates a gap in the record the Commission must resolve before acting. This is not a question of outcome. It is a question of process. If the conditions requiring evaluation are present, the evaluation must exist. If the evaluation exists, it must be reflected or accounted for where it bears on representations made to the Commission. If neither appears on the record, the Commission cannot assume compliance as a predicate to action.

The ITAR compliance question and the Section 1.65 ownership question are not separate concerns. They are parallel compliance regimes addressing the same risk vector: foreign access to controlled systems operated by a regulated entity. ITAR requires evaluation when foreign persons access controlled facilities. Section 1.65 requires disclosure when foreign interests enter a regulated entity's ownership structure. Executive Order 13913 and 47 C.F.R. Sections 1.5000 et seq. require the Commission to coordinate with the Executive Branch when foreign participation raises national security concerns. The record presents unresolved questions under all four frameworks simultaneously, including the Opposition's execution posture under Section 1.52. The Applicant's Opposition addressed none of them.

The identity of the individuals involved is not material to the Commission's analysis. The relevance is limited to the regulatory condition: the documented presence or attempted presence of foreign persons at a controlled facility, operated by the same entity now seeking the largest

satellite authorization in Commission history, combined with a post-merger ownership structure that introduces foreign sovereign-affiliated capital the Commission has not evaluated. The Commission need not resolve the underlying ITAR question to recognize that these parallel compliance gaps, taken together, prevent the affirmative public interest finding Section 309(a) requires.

D.10 Preservation

Petitioner preserves all arguments arising from the ITAR disclosure question to the fullest extent allowable by law, including but not limited to their application to the Commission's character and candor evaluation, the weight assigned to the Applicant's unsworn representations regarding compliance history, and the public interest determination under 47 U.S.C. Section 309(a).

APPENDIX E: ODC PROLIFERATION, DELEGATED AUTHORITY, SUNSHINE ACT, AND THE FORUM THIS PROCEEDING REQUIRES

This Appendix addresses the institutional and procedural framework within which this Application should be evaluated. Multiple orbital data center applications are now emerging before the Commission. The Commission's own Part 100 NPRM acknowledges the Part 25 framework was not designed for these systems. Whatever the Commission does with this Application establishes precedent for every application that follows.

E.1 The Novelty Requires Commission-Level Deliberation

At least four independent novel questions are presented: (1) whether Part 25 authorizes licensing of a system whose primary function is orbital AI computing; (2) whether an environmental certification can be sustained on a record containing no aggregate atmospheric data against peer-reviewed science; (3) whether the cumulative waiver package can be evaluated under existing precedent; and (4) whether the ownership disclosure satisfies mandatory requirements where the post-merger ownership picture is not before the Commission. Each of these is a first-impression question. Under the Commission's own delegation framework, such questions should be referred to the Commission en banc. See Appendix A, Section I.

E.2 The Sunshine Act Governs the Form

If any disposition involves the deliberation of a quorum of Commissioners, the Government in the Sunshine Act requires public deliberation. The proceeding is not routine. The Chairman's public statements during the pleading cycle, cited by the Applicant as authority, and documented Commissioner engagement with the Application's branding raise the concerns the Sunshine Act was designed to address. See Appendix A, Section II.

E.3 Section 309(e) Mandatory Hearing Threshold

Where the Commission cannot make the affirmative public interest finding, or where a substantial and material question of fact is raised, Section 309(e) commands hearing designation. The statute uses "shall." The ownership question, the classification question, the character question, and the environmental question each independently approach this threshold. Together, they satisfy it. See Appendix A, Section III.

E.4 A Civilizational System Deserves a Proportionate Proceeding

The Applicant described its system as a first step toward a Kardashev II-level civilization. If the system is as consequential as the Applicant's own narrative claims, then the proceeding that evaluates it should be commensurate with that consequence. A Bureau-level grant on delegated authority, on an unsworn record that does not conform with Section 1.52, with an unresolved ownership disclosure, a contradicted environmental certification, and documented character questions across four independent compliance regimes, does not meet that standard. The Commission should identify the forum this record requires and proceed accordingly.

E.5 The Chairman's Public Statement and the Integrity of the Adjudicative Process.

At footnote 5 of the Opposition, the Applicant cites Chairman Carr's March 11, 2026 post on X as authority for the proposition that a competing petitioner should not have exercised its statutory rights under 47 U.S.C. Section 309(d). The Applicant offers this citation in lieu of a substantive response on the merits of that petitioner's filing.

Petitioner takes no position on the merits of the filing the Chairman's post addressed. The concern here is structural, not substantive.

The Commission's adjudicative proceedings are subject to due process requirements that include the obligation of impartiality on the part of decision-makers. When the head of an agency publicly characterizes the exercise of a statutory right as improper during an open pleading cycle in which that agency official will participate in the final disposition, the question of prejudgment arises. The D.C. Circuit has held that agency decision-makers in adjudicatory proceedings must maintain the appearance of impartiality, and that public statements indicating a closed mind on a pending matter may constitute a due process violation requiring disqualification. See *Cinderella Career & Finishing Schools, Inc. v. FTC*, 425 F.2d 583, 590 (D.C. Cir. 1970). The D.C. Circuit has separately held that off-the-record communications in proceedings required to be decided on the record may taint the proceeding and require vacatur. See *Sangamon Valley Television Corp. v. United States*, 269 F.2d 221 (D.C. Cir. 1959).

The concern is compounded when the regulated party treats the decision-maker's public statement not merely as political commentary but as a component of its own legal argument. By citing the Chairman's post in its Opposition, the Applicant incorporated the statement into the formal record of this proceeding. The Commission's Office of General Counsel and the Office of Government Ethics, under 5 C.F.R. Part 2635, are the appropriate bodies to evaluate whether the

Chairman's participation in the disposition of this Application is consistent with federal standards of ethical conduct for employees of the executive branch, including the requirement under 5 C.F.R. Section 2635.502 that employees not participate in matters where their impartiality could reasonably be questioned.

Petitioner does not assert that the Chairman has committed an ethical violation. Petitioner observes that the Applicant's own filing has created a record in which the question must be evaluated. To the extent the Commission's Office of General Counsel determines that referral to the Office of Government Ethics for an advisory opinion is warranted, Petitioner respectfully requests that any such review be completed before final disposition of this Application. A final agency action in which a decision-maker's impartiality is subject to unresolved question inherits a reviewable defect that no amount of substantive analysis can cure.

Petitioner preserves all arguments arising from the Chairman's public statement, the Applicant's citation of that statement, and the Commission's adjudicative obligations under the Due Process Clause, the APA, and the authorities cited above, for reconsideration under 47 U.S.C. Section 405 and judicial review under 47 U.S.C. Section 402(b) and 5 U.S.C. Section 706.

APPENDIX F: RECORD SUFFICIENCY MATRIX (TWENTY-TWO OBSERVATIONS)

The Bakken SIO (March 15, 2026) condensed the record into twenty-two structural observations. Each observation identified a specific discrepancy, gap, or inconsistency in the Application's certified record and closed with the same fork: adjudicate on the record as filed, or acknowledge that any cure constitutes a material amendment requiring new public notice under 47 C.F.R. Section 25.116.

The Consolidated Opposition did not address any of the twenty-two observations with curative amendments, corrected filings, or factual rebuttal. Each therefore remains uncontested on the record and is preserved for all purposes including petition for reconsideration under 47 U.S.C. Section 405 and judicial review under 5 U.S.C. Section 706.

The full text of the Bakken SIO is incorporated by reference. The twenty-two observations address: (1) the Schedule S encoding discrepancy; (2) the epoch date inconsistency; (3) the validation error; (4) the absence of per-satellite mass data; (5) the absence of area-to-mass ratio; (6) the absence of DAS calculations at scale; (7) the absence of passive-decay analysis; (8) the absence of conjunction screening methodology; (9) the stale Ownership Attachment; (10) the absence of board resolution for signatory authority; (11) the service classification omission; (12) the absence of ITU coordination data; (13) the waiver specificity deficit; (14) the environmental certification gap; (15) the absence of annualized replacement rates; (16) the absence of

disposal-method split data; (17) the infrastructure dependency on Gen2 ISL capacity; (18) the regulatory classification ambiguity; (19) the cross-docket consistency questions; (20) the pre-merger filing timing; (21) the acceptance timeline during government shutdown; and (22) the cumulative materiality of the foregoing.

Each of these observations is independently preserved and independently operative.

APPENDIX G: THE SCHEDULE S WAIVER AS BLANKET AUTHORITY REQUEST

The Application requests waiver of 47 C.F.R. Section 25.114(a)(1), the Schedule S filing requirement. The Applicant's waiver narrative states that the form cannot accurately describe the proposed system. This Appendix addresses the nature of that waiver request and its implications for the Commission's processing framework.

G.1 The Hybrid Character of the Waiver

The Application's non-Schedule-S waivers (processing rounds under Section 25.156(d)/25.157, milestones under Section 25.164, surety bond under Section 25.165) are specific: they identify the rule, state the basis, and request relief from a discrete obligation. They are substantive waivers of discrete requirements, each of which can be individually evaluated under the good cause standard of 47 C.F.R. Section 1.3.

The Schedule S waiver is different. Schedule S is not a single rule. It is a structured filing instrument that encodes orbital parameters, frequency plans, antenna patterns, and system architecture into a machine-readable format the Commission uses for interference analysis, coordination screening, and processing. Waiving Schedule S does not waive a single rule. It waives the structured data layer that permits the Commission to evaluate the application against virtually every other technical rule in Part 25.

G.2 Why This Functions as a Blanket Authority Request

The Commission's 2023 Final Rule (FCC 23-73) explicitly prohibits blanket waivers. An applicant cannot satisfy Section 25.112(b)(1) with a blanket request for waiver of any unspecified rule. The Schedule S waiver, as framed in this Application, functions as precisely that kind of request. By stating that the form cannot describe the system, the Applicant is effectively requesting that the Commission accept an application it cannot technically evaluate using its standard tools, on the representation that the system will comply with rules the Commission cannot verify on the record before it.

That is not a waiver of a specific rule for a specific reason. It is a request for blanket authority to operate outside the structured framework the Commission's rules establish for evaluation, processing, and coordination of NGSO satellite systems.

G.3 The Certification Paradox

The Form 312 certification, executed under 18 U.S.C. Section 1001, states that the information provided is true, complete, and correct. The waiver narrative states the form cannot accurately describe the system. Both statements are on the same record. Both bear the same certifier's signature. They cannot both be true. The Commission must resolve this paradox on the record before it can evaluate the technical merits of the Application.

Appendix H: The Commission's Own Framework Applied to the ODC Application

H.1. Incremental Authorization and Scale.

The Commission authorized 7,500 of 29,988 requested Gen2 satellites, not the full constellation, specifically because of scale concerns. FCC 22-91 at para. 19: the "smaller number of satellites will allow continued monitoring of deployment based on conditions adopted in this Order, prior to consideration of the much larger number of satellites SpaceX requests over the long term." The ODC Application requests authorization for up to one million satellites, more than thirty-three times the full Gen2 constellation the Commission chose not to authorize in whole, on a record that contains fewer bounded parameters than the Gen2 application provided.

H.2. Physical Characteristics and Technical Data.

Before granting Gen2, the Commission required the Applicant to provide physical satellite characteristics and sample DAS logs sufficient for the Commission to complete its review. FCC 22-91 at para. 16. The Applicant provided antenna beam patterns, physical dimensions, mass data, and collision risk calculations for each proposed orbital shell. The ODC Application contains no per-satellite mass, no physical dimensions, no disposal-method split, no annualized replacement rate, no aggregate reentry mass, no DAS calculations, and no conjunction screening at fleet scale. The Opposition states the Applicant "is conducting" and "will conduct" these analyses. Opposition at 8. The Commission did not accept that posture for 29,988 satellites. It required bounded data on the record before grant.

H.3. NASA Coordination and the Opposition's Mischaracterization.

The Commission conditioned Gen2 deployment below the ISS on completed physical coordination with NASA and limited initial deployment to 400 satellites until further coordination was documented. DA 24-1193 at para. 2.

The Opposition cites "NASA Comments" at pages 4, 5, 7, 11, 13, 15, and 20, characterizing NASA's input as collaborative engagement supporting the Application's progress toward authorization. The actual NASA letter filed in this docket does not support that characterization.

NASA's letter, filed under ICFS File No. SAT-LOA-20260108-00016, is the Letter from Lauren E. Morgan, NASA Representative to the Commercial Transportation Interagency Group, Space Operations Mission Directorate, Launch Services Office, to Marlene H. Dortch, Secretary, FCC. The letter independently confirms the following conditions on this record:

Milestone waiver risk. NASA states that the Applicant's request for waiver of Section 25.164 milestone requirements creates risks for NASA because "there is limited information about the

system such as its orbital shells, the number of satellites in each shell, and the timelines for deployment and disposal." NASA states this uncertainty "could significantly affect NASA's ability to plan future mission deployments and operations."

Absence of bounded orbital parameters. NASA states that "[w]ithout detailed information on the proposed system's orbital parameters particularly given the potential scale of up to one million satellites in low Earth orbit NASA cannot accurately assess collision risks."

Disposal plan deficiency. NASA states that "it is unclear whether such plans are scalable and sustainable for the proposed system" and that "there are minimal details on how many satellites will be disposed via atmospheric reentry, how many will be raised above 2000km for disposal, and when such transitions will occur across the lifetime of the fleet." NASA further states that the Applicant's attestation regarding recycling and reuse "cannot be considered viable given the lack of knowledge as to whether such activities are feasible for such a large system."

Collision arithmetic. NASA calculates that based on the Applicant's own stated disposal parameters, "the number of collisions among up to 1 million retired spacecraft in disposal orbits above 2,000 km is up to 10,000 for 100 years. An average of up to 100 collisions per year, which can lead to serious permanent negative damage to the environment."

Failure rate arithmetic. NASA states that even at 99.9% reliability, "it still means 1,000 spacecraft can potentially fail, due to small MMOD impacts, at their operational orbits." NASA states that "[f]ragmentation debris from mutual collisions among failed spacecraft in narrow operational shells can have serious negative effects on other operators' active spacecraft in LEO."

Reentry casualty arithmetic. NASA observes that the Applicant's stated casualty threshold of less than $1e-4$ per reentry, applied to one million spacecraft, "can lead to an expectation of casualty of 100 from the reentries of 1 million spacecraft." NASA recommends that "[f]ull demise of the spacecraft during reentry should be required for the constellation."

Degradation debris. NASA states that "[t]he total surface area, including large solar arrays, from a constellation of 1 million spacecraft will become the most dominant source of degradation debris in LEO" and that "[t]he negative effect of the generation of mission-critical degradation debris from SpaceX's Orbital Data Center and the risk increase to other operational spacecraft in LEO need to be carefully evaluated."

ISS coordination. NASA requests clarification of all deployment and phasing orbits and coordination with NASA when planned maneuvers intersect the ISS altitude band of approximately 400 to 430 km. NASA notes that the Application indicates a shell perigee of 500 km with a tolerance of plus or minus 100 km, meaning perigee could be as low as approximately 400 km, intersecting the ISS altitude band. NASA requests either tightened perigee tolerance, raised perigee, or documented coordination for any crossings.

Laser interference with science missions. NASA states that the proposed optical inter-satellite links "could potentially interfere with highly sensitive space-based instruments and star trackers" and requests information on laser divergence angle, risk of direct beam impingement, and the possibility of reflections that could scatter energy toward sensitive detectors.

Launch collision avoidance. NASA states that given the range and ambiguity of orbital parameters, "launch windows may be negatively impacted" and that "[f]urther information on the orbital parameters and deployment schedule for the proposed system is needed."

In every material respect, NASA's letter supports the deficiencies Petitioner identified and contradicts the Opposition's characterization of NASA's position. NASA's letter reads not as the collaborative endorsement the Opposition portrays, but as an independent identification of the same categories of missing data, unresolved risk, and insufficient technical basis that Petitioner's filings documented.

The Opposition confirms every response to NASA was a commitment to future coordination, not the provision of present data. Opposition at 4, 11, 13, 15, 20. The Commission conditioned 7,500 Gen2 satellites on NASA coordination and bounded technical data on the record before grant. The ODC record contains no documented responsive data for a system more than one hundred times that size.

The distance between what the Opposition represented regarding NASA's position and what NASA actually stated in its own filing is material to the Commission's evaluation of the Opposition's reliability, particularly in a document that does not conform with 47 C.F.R. Section 1.52.

H.3.1. The International Astronomical Union Filing.

The International Astronomical Union (IAU), representing more than 12,000 professional astronomers from approximately 90 countries, filed comments in this docket stating that the proposed system of up to one million ODCs is "incompatible with preserving dark and quiet skies for science and society." The IAU stated that even deployment at 10% of the proposed scale, 100,000 satellites, "would severely interfere with scientific and other uses of orbital space and the sky."

The IAU's analysis independently confirms several conditions relevant to the Commission's evaluation:

Optical brightness. The IAU estimates that ODC satellites will greatly exceed the IAU-recommended brightness threshold, with visual magnitudes commonly ranging between 4 and 1. The IAU anticipates more visible satellites in the sky than visible stars for many latitudes, times of night, and times of year.

Impacts on U.S. observatories. The IAU identifies specific risks to twilight programs including planetary defense efforts, wide-field surveys such as those conducted by the Vera C. Rubin Observatory, and societal impacts from severely light-polluted skies.

Infrared, submillimeter, and radio interference. The IAU anticipates consequences for astronomy at these wavelengths due to satellite thermal radiation and unintended electromagnetic radiation (UEMR) from electronic noise, which "could lead to major data loss for low-frequency radio observatories throughout the world, including US facilities."

Debris generation from meteoroid impacts. The IAU estimates, using the NASA meteoroid environment model, over 30 million meteoroid collisions per year with the constellation's aggregate surface area, which "could substantially add to the small debris population."

Atmospheric effects. The IAU estimates the system would require approximately 10 to 20 million tonnes of fuel per year, much of which would be deposited into the stratosphere and mesosphere, constituting "a substantial change from current rocket launches."

Space-based observatories. The IAU emphasizes that interference extends to space telescopes due to the need to observe through shells of satellites.

The IAU recommends that any license be conditioned on coordination with the National Science Foundation and the American Astronomical Society, and expresses "serious concern about the use of waivers that could avoid normal checks on the placement of satellite systems into orbit."

The IAU filing was signed by its President, Prof. Willy Benz, and represents the position of the international scientific community. It independently corroborates the environmental, technical, and observational deficiencies identified by Petitioner, NASA, and other parties on this record. The Opposition did not address the IAU's concerns.

H.4. Orbital Debris Failure Thresholds.

The Commission imposed a quantitative failure threshold on Gen2: if cumulative post-failure object years exceeded 100, the Applicant was required to cease deployment until the Commission approved an updated orbital debris mitigation plan. FCC 22-91 at para. 85. The ODC Application contains no failure rate analysis, no post-failure object-year projections, and no proposed threshold for pausing deployment. The Commission's own framework requires this analysis before grant.

H.5. EPFD Analysis and ITU Coordination.

The Commission required the Applicant to submit a single combined EPFD data file for all Gen2 satellites, to make that file available to requesting parties, and to obtain a favorable or qualified favorable ITU finding. FCC 22-91 at paras. 31-34. The ODC Application contains no EPFD

analysis, no combined data file, and no ITU advance publication or coordination data. The Commission's own immediately prior order for this same Applicant required this information as a precondition.

H.6. Schedule S Waiver Precedent.

The Commission granted a Schedule S waiver for Gen2 only because the Applicant had "implemented workarounds to provide the relevant information" including "complete frequency use and orbital information for its constellation in separate databases." FCC 22-91 at para. 25; DA 24-1193 at para. 84. The ODC Application's Schedule S encodes three satellites for a system described as one million. The Opposition acknowledges the form cannot describe the system. No supplemental data has been filed as a workaround. No amendment has been submitted. The Opposition introduces a representative orbital configuration in its Table 1 containing 998,240 satellites, but that information does not appear in the Application and was not filed through the procedural channels the Commission's own precedent contemplates.

H.7. Environmental Review and Scale-Dependent Reasoning.

The Commission declined to require environmental review for Gen2 in part because of the limited number of satellites authorized. FCC 22-91 at para. 117: "We note that we are not currently authorizing SpaceX to deploy all of its proposed 29,988 satellites, but rather a smaller subset of 7,500 satellites... Therefore, the parties' estimated amount of alumina that could be introduced into the atmosphere will not come to pass from our action today." The Commission also relied on two ESA studies finding atmospheric impact from reentering spacecraft to be negligible, but those studies assumed a worst case of 450 to 650 tons of reentering satellites per year. FCC 22-91 at para. 118. One million ODC satellites with a five-year design life implies approximately 200,000 atmospheric reentries per year at steady state. The assumptions underlying the Commission's prior environmental determination are exceeded by the ODC Application by orders of magnitude. To rely on the same reasoning here, the Commission would need to explain why the same logic applies at a fundamentally different scale. NASA's own letter in this docket calculates that the Applicant's stated casualty threshold, applied to one million spacecraft, produces an expected casualty count of 100. The IAU independently estimates approximately 10 to 20 million tonnes of fuel deposited into the stratosphere and mesosphere per year. These are not Petitioner's calculations. They are the calculations of the agencies and international scientific bodies the Commission's own framework relies upon.

H.8. FAA Reliance Gap.

The Commission relied on FAA environmental assessments covering specific launch vehicle cadences from specific facilities. FCC 22-91 at para. 115. Those assessments cover a combined cadence of approximately 145 launches per year across three sites. The ODC Application does not state how many launches the proposed system requires, but the implied scale suggests a

launch cadence far exceeding what any existing FAA environmental assessment has evaluated. The Commission's own reasoning requires an FAA review to exist before the Commission can rely on it. For the ODC system, the record does not identify one.

H.9. Cross-Docket Infrastructure Dependency.

The Commission treated Gen1 and Gen2 as separate systems for processing round, EPFD, milestone, and bond purposes, while acknowledging they share user terminals and spectrum. FCC 22-91 at para. 46. The ODC system relies on Gen2's optical inter-satellite link infrastructure for connectivity to the Starlink network. The record does not contain any cross-docket analysis of how the ODC's dependence on Gen2 infrastructure affects Gen2's existing conditions, its authorized capacity, or its compliance obligations. The Commission has not evaluated whether granting the ODC Application would effectively modify the terms under which Gen2 operates, without following the modification procedures the Commission's own rules require.

H.10. Standard of Review.

Each of these comparisons is drawn from the Commission's own published orders, applied to the same Applicant, for the same category of system. The standard is not Petitioner's invention. It is the Commission's established framework, applied consistently, at the scale the Application itself proposes.

APPENDIX I: WAIVER AND PROCEDURAL SURVIVABILITY

This Appendix consolidates Petitioner's procedural waiver requests, severability provisions, and related preservation clauses. It is incorporated by reference in the body of the Reply and operates independently.

I.1 Filing Posture

Petitioner appears pro se in this proceeding and is not represented by counsel. This Reply is submitted within the reply window of the pleading cycle established by Public Notice DA 26-113, in a form and through a procedural vehicle Petitioner believes to be proper.

I.2 Procedural Compliance and Waiver

The Public Notice governing this proceeding directed participants to file "in accordance with section 25.154 of the Commission's rules." That is the only procedural rule the Public Notice identified. Petitioner has filed in accordance with 47 C.F.R. Section 25.154.

To the extent this Reply satisfies Section 25.154 but is deemed deficient under any other provision of the Commission's rules, Petitioner requests waiver of that provision pursuant to 47 C.F.R. Section 1.3. To the extent this Reply is deemed deficient under Section 25.154 itself, Petitioner requests waiver of the specific requirement not satisfied, also pursuant to Section 1.3. To the extent additional procedural requirements apply that were not identified in the Public Notice, the Commission's own filing infrastructure did not provide the notice necessary for an unrepresented participant to identify them. Pro se filings are held to a less stringent standard than those drafted by attorneys. *Haines v. Kerner*, 404 U.S. 519, 520 (1972).

Specifically, to the extent required, Petitioner requests waiver of:

(a) 47 C.F.R. Section 1.52, to the extent the Commission determines that this Reply requires verification beyond the signature provided. The factual allegations supporting this Reply are contained in the sworn declarations filed with the Bakken EP and Bakken IR, both incorporated by reference.

(b) 47 C.F.R. Section 25.154(a)(5), to the extent the Commission determines that service by electronic mail does not satisfy the requirement that a pleading be "mailed to the applicant." The Commission granted waiver of this same provision under comparable circumstances in the Gen2 Upgrade proceeding. DA 24-1193 at fn. 163.

In the alternative, Petitioner requests leave to cure any identified deficiency before any adverse action is taken.

To the extent any waiver request in this section is itself subject to a procedural requirement Petitioner has not identified or satisfied, Petitioner requests waiver of that requirement on the same grounds stated above. Silent application of an undisclosed procedural requirement to exclude this filing from the record would constitute the kind of undisclosed legal standard the D.C. Circuit has held incompatible with reasoned agency action. *NLRB v. Sears, Roebuck and Co.*, 421 U.S. 132 (1975); 5 U.S.C. Section 552(a)(2).

Nothing in this waiver request concedes that any waiver is required. It is filed in an abundance of caution to ensure that procedural form does not prevent the Commission from evaluating a filing that bears directly on the condition of the record before it.

I.3 The Asymmetry Standard

The Commission cannot deny Petitioner's procedural waiver while accepting the Applicant's substantive waivers of the processing round, milestone, surety bond, and Schedule S requirements that constitute the framework for evaluating applications of this scale. If the Commission concludes that good cause supports waiver of the substantive rules the Applicant's filing does not satisfy, it cannot simultaneously conclude that good cause does not support waiver of procedural rules the Commission's own infrastructure made difficult for an unrepresented participant to identify. The standard under Section 1.3 is the same. A proceeding in which the Applicant's certified deficiencies are accommodated through waiver while the objector's procedural imperfections are treated as fatal would not satisfy the Commission's obligation to act in a non-arbitrary manner. *See* Section V.A of this Reply (Prayer for Relief).

I.4 Execution and Verification of the Consolidated Opposition

47 C.F.R. Section 1.52 distinguishes between filings submitted by counsel and those submitted without representation. The Consolidated Opposition was signed by a Senior Satellite Policy Manager. The signatory was not identified as an attorney of record. The Opposition is unsworn. It does not bear the signature of a corporate officer, an engineering officer, or the certifying official who executed the Form 312.

This posture informs the weight the Commission may assign to the representations contained therein, particularly where they are offered to address disputed factual issues. That principle applies equally across all filings in this proceeding. The full analysis of the Opposition's conformity with Section 1.52, including the entity mismatch between the signatory's employer (Space Exploration Technologies Corp.) and the applicant entity (Space Exploration Holdings, LLC), the absence of demonstrated authorization under Sections 1.21(d) and 1.22, the mandatory verification fork, the absence of counsel from the document while HWG LLP was served by at least two parties in this proceeding, and the seven-day cure demand with striking remedy, is set forth at Section I of this Reply. This Reply is accompanied by a Verification under penalty of perjury executed by Petitioner.

I.5 Severability

Each section of this Reply, each argument, each identification of a record deficiency, and each request for relief is independently stated and independently preserved. A procedural defect in one portion does not extinguish the Commission's obligation to evaluate the remaining portions. To the extent any portion requires reclassification to a different procedural vehicle in order to be received, Petitioner requests that the Commission reclassify rather than reject.

I.6 Preservation of FOIA-Dependent Arguments

Petitioner has submitted expedited Freedom of Information Act requests with the FCC as documented in the Bakken IR. Those requests seek records bearing directly on the acceptance determination, internal processing standards, and the Commission's pre-filing communications regarding this Application. To the extent those records, when produced, reveal information that would have materially informed this Reply or any prior filing, all arguments arising from that information are preserved.

I.7 Constitutional Protection and Incorporation

This Reply, and each prior filing in this proceeding, constitutes a petition for redress of grievances protected by the First Amendment and the Administrative Procedure Act. 5 U.S.C. Section 555(b); *Borough of Duryea v. Guarnieri*, 564 U.S. 379, 388 (2011). All prior submissions are incorporated herein by reference in their entirety, remain independently operative, and are not narrowed, waived, or superseded by anything in this filing.

APPENDIX J: THE TERAFAF DISCLOSURE AND THE RECORD THE APPLICANT CHOSE NOT TO FILE

J.1. What Occurred.

On the evening of March 21, 2026, the Applicant's controlling principal held a public event at the Seaholm Historic Power Plant in Austin, Texas. The event was jointly presented by SpaceX, Tesla, and xAI. It was livestreamed. It was attended by the Governor of Texas. It was covered in real time by SpaceNews, Electrek, Teslarati, KUT (Austin's NPR affiliate), the Taipei Times, CGTN, and numerous other outlets. SpaceX and Tesla simultaneously posted promotional content on X, the platform owned by the Applicant's corporate parent. See Jeff Foust, "SpaceX offers details on orbital data center satellites," SpaceNews (Mar. 22, 2026); Fred Lambert, "Tesla and SpaceX announce \$25B 'Terafab' chip factory," Electrek (Mar. 22, 2026); KUT Radio, "Elon Musk announces chip manufacturing facility in Austin" (Mar. 22, 2026).

The event occurred one day before the close of the reply window in this proceeding. No party had the opportunity to evaluate, respond to, or file comments on the information disclosed. The Applicant has not filed any of the information presented at this event with the Commission.

J.2. The Physical Satellite Design: "AI Sat Mini."

The Applicant displayed an illustration of the satellite that would initially compose the Orbital Data Center constellation, designated the "AI Sat Mini." The illustration was described by the Applicant's controlling principal as drawn to scale alongside the Starship V3 launch vehicle. The Starship V3 is 124 meters tall.

The following physical parameters were disclosed:

Mass: approximately one metric ton per satellite. See NotATeslaApp, "Everything Announced at Tesla's Terafab Event" (Mar. 22, 2026), reporting: "Each AI Sat Mini should be approximately 1 ton."

Physical dimensions: When unfolded, including solar panels and radiators, the satellite is wider than the Starship V3 is tall. See SpaceNews (Mar. 22, 2026), reporting: "If the spacecraft was shown at the same scale as Starship, it would be more than 170 meters long." See also Steve Jurvetson (SpaceX board member), post on X (Mar. 21, 2026): "First view of the 100kw AI Mini Sat with solar panels and heat radiator to scale."

Each individual Orbital Data Center satellite, at more than 170 meters in length when deployed, is larger than the rocket that launches it.

Solar arrays: The satellite design is "dominated by large solar arrays." SpaceNews (Mar. 22, 2026). The solar arrays provide 100 kilowatts of power for onboard AI processors. Id.

Radiator: The satellite features a radiator covering approximately 100 square meters for thermal management. SpaceNews (Mar. 22, 2026).

Power: 100 kilowatts per satellite, with the Applicant's controlling principal stating that future satellites are expected to scale to the "megawatt range." SpaceNews (Mar. 22, 2026); NotATeslaApp (Mar. 22, 2026).

None of these parameters appear in the Application filed with the Commission. The Schedule S encodes three satellites with no physical dimensions, no mass data, no power specifications, no solar array area, and no radiator area. The Form 312 certifies the Application as true, complete, and correct under 18 U.S.C. Section 1001. Nine threshold defects identified in Table 1 of this Reply include the absence of per-satellite mass, physical dimensions, and bounded technical parameters at fleet scale. The Applicant possesses this data. It presented it to the public. It has not filed it with the Commission.

J.3. The Scale Implied by the Disclosed Parameters.

One million satellites at approximately one metric ton each implies an aggregate orbital mass of approximately one million metric tons. One million satellites with a five-year design life implies approximately 200,000 atmospheric reentries per year at steady state, each depositing the combustion byproducts of a one-ton satellite into the stratosphere. The Applicant's own presentation stated that the system would require launching "10 million tons of mass to orbit per year" at full scale. See Cryptobriefing, "Elon Musk unveils Terafab in bid to unlock massive AI compute in space" (Mar. 22, 2026). The Applicant's controlling principal described a throughput of "100 million tons of solar capture equipment into space every year" as a long-term objective. TeslaNorth (Mar. 21, 2026).

The environmental certification on the Commission's record contains no aggregate mass data, no annualized reentry tonnage, no atmospheric chemistry modeling, and no assessment of the lifecycle throughput these figures imply. The Bakken EP, NASA's letter, the IAU's filing, and DarkSky International's filing each independently identified the absence of this data. The Applicant now confirms it has parameters to disclose. It disclosed them to the press, not the Commission.

J.4. The Custom Chip: "D3."

The Applicant disclosed that a custom semiconductor, designated the "D3," has been designed specifically for the Orbital Data Center satellites. The D3 chip is radiation-hardened for the space environment and designed to operate at elevated temperatures to minimize the mass of thermal radiators. See SpaceNews (Mar. 22, 2026), reporting that the chip is "optimized for use in space, designed to run hotter than terrestrial chips and with radiation protection." See also Cryptobriefing (Mar. 22, 2026), describing the chip as "purpose-built for space conditions, engineered to withstand high-energy particle bombardment."

The Commission's record does not reference the D3 chip. It does not contain any information regarding the computing hardware, radiation hardening approach, thermal operating parameters, or power consumption characteristics of the satellites the Application proposes to deploy.

J.5. The Chip Fabrication Program: "Terafab."

The Applicant announced a joint venture among SpaceX, Tesla, and xAI to construct a semiconductor fabrication facility designated "Terafab" at the North Campus of Giga Texas in Austin, Texas. The following details were disclosed:

Estimated cost: \$20-25 billion. See Electrek (Mar. 22, 2026). Tesla's Chief Financial Officer acknowledged at the event that the full Terafab cost is not yet incorporated into Tesla's capital expenditure plan for 2026. Id.

Production target: one terawatt of AI computing power per year, described as approximately 50 times the current combined global production of AI-class chips. SpaceNews (Mar. 22, 2026).

Allocation: 80% of Terafab's compute output directed to space-based orbital AI satellites, 20% for ground-based applications. Electrek (Mar. 22, 2026); Teslarati (Mar. 22, 2026); New Space Economy (Mar. 22, 2026).

Technology node: targeting 2-nanometer process technology. Electrek (Mar. 22, 2026).

Initial production: 100,000 wafer starts per month, with ambitions to scale to 1,000,000 wafer starts per month at full capacity, representing approximately 70% of TSMC's current global output. Electrek (Mar. 22, 2026).

Chip output: between 100 and 200 billion custom AI and memory chips per year. Electrek (Mar. 22, 2026).

Scope: the facility will consolidate chip design, lithography, fabrication, memory production, advanced packaging, and testing under one roof. SpaceNews (Mar. 22, 2026); KUT Radio (Mar. 22, 2026).

The Commission's record does not reference Terafab. The Application does not describe the chip fabrication program, the \$20-25 billion capital commitment, the corporate structure of the joint venture, or the allocation of output between terrestrial and space applications. The Applicant's controlling principal stated at the event: "We either build the Terafab or we don't have the chips, and we need the chips, so we're going to build the Terafab." SpaceNews (Mar. 22, 2026). The chips are described as the "missing ingredient" in the Applicant's plans. Id.

J.6. The Corporate Structure Disclosed at the Event.

The Terafab event was presented as a joint initiative of three entities: SpaceX, Tesla, and xAI. SpaceX and Tesla simultaneously posted promotional materials on X. The Governor of Texas was in attendance. KUT Radio (Mar. 22, 2026).

The Ownership Attachment on the Commission's record predates the xAI merger (completed February 2, 2026). It does not reflect the Tesla relationship. It has not been updated. The 30-day self-executing amendment deadline under 47 C.F.R. Section 1.65 expired March 4, 2026. As of the date of this Reply, no amended disclosure has been filed. The Terafab announcement confirms that the Applicant's corporate structure has changed in ways directly relevant to the Application and that the Applicant is actively presenting that changed structure to the public while declining to disclose it to the Commission.

J.7. The Scaling Roadmap.

The Applicant's controlling principal disclosed a scaling roadmap for the Orbital Data Center system that extends well beyond the parameters in the Application:

Future satellites scaling from 100 kilowatts to the "megawatt range." SpaceNews (Mar. 22, 2026).

A requirement for "10 million tons of mass launched to orbit per year." NotATeslaApp (Mar. 22, 2026); Cryptobriefing (Mar. 22, 2026).

An eventual objective of "100 million tons of solar capture equipment into space every year." TeslaNorth (Mar. 21, 2026).

One billion Optimus robots for physical manufacturing. NotATeslaApp (Mar. 22, 2026).

Electromagnetic mass drivers on the lunar surface to launch payloads without rocket propellant. SpaceX post on X (Mar. 22, 2026); NotATeslaApp (Mar. 22, 2026); New Space Economy (Mar. 22, 2026).

The Applicant's controlling principal framed the project as achieving "a galactic civilization" and referenced the Kardashev Scale, the same civilizational framing the Application uses in its Narrative but the Opposition did not mention once in thirty-two pages. The presentation described an "era of Amazing Abundance" in which humanity would travel "beyond the Moon, beyond Mars, and through the rings of Saturn." NotATeslaApp (Mar. 22, 2026); KUT Radio (Mar. 22, 2026).

None of this scaling roadmap is on the Commission's record. None of it was available to any party during the pleading window.

J.8. The Record Implications.

The March 21, 2026 disclosures bear on this proceeding in at least six independent respects.

First, completeness and the Section 1001 certification. The Applicant possesses bounded physical satellite parameters, including mass, dimensions, power, radiator area, and solar array configuration. The Form 312 certifies the Application as true, complete, and correct. The Application contains none of these parameters. The Applicant's decision to disclose them to the public while the certified record remains bare is relevant to the Commission's evaluation of the certification's continuing accuracy.

Second, the environmental certification. One million satellites at one metric ton each, with the Applicant's own stated throughput of 10 million tons of mass to orbit per year, implies an atmospheric loading regime that the Commission's record does not evaluate, that the Applicant's environmental certification states has no significant impact, and that the Applicant's own

Opposition admits requires real-world deployment to assess. The scale now publicly confirmed by the Applicant exceeds the assumptions underlying the Commission's Gen2 environmental determination by orders of magnitude.

Third, the Section 1.65 ownership obligation. The Terafab announcement confirms the operational integration of SpaceX, Tesla, and xAI. The Ownership Attachment on the Commission's record predates the xAI merger and does not reflect the Tesla relationship. To the extent the Terafab corporate structure constitutes a material change in the relationships among entities bearing on the ownership and control of the Applicant, the Section 1.65 amendment obligation is independently triggered.

Fourth, the Opposition's technical representations. The Opposition introduced Table 1 (998,240 satellites) as an unsworn, uncertified document that does not appear in the Application. The Terafab event disclosed additional technical parameters, including per-satellite mass, power, and physical dimensions, that also do not appear in the Application and were not available to any party during the comment or opposition windows. The Commission should determine whether it will evaluate the Application on the certified record before it, or on representations the Applicant has made to the public but has not filed.

Fifth, character and candor. An applicant that requests "expeditious" grant while simultaneously withholding from the Commission the technical data it is presenting to the press, investors, and the public raises the concerns the Commission's 1986 Character Qualifications Policy, 102 F.C.C.2d 1179, exists to address. The Commission relies on licensee honesty in a largely self-policing system. *Contemporary Media, Inc. v. FCC*, 214 F.3d 187, 193 (D.C. Cir. 2000). An applicant that presents bounded technical parameters at a press event the night before the record closes, while the Commission's own record encodes three satellites with no physical data, is not cooperating with the regulatory process.

Sixth, the pleading window and due process. The technical details disclosed on March 21, 2026 were not available to any party during the opposition window (which closed March 6) or during the reply window. No party had the opportunity to evaluate the satellite dimensions, mass, power architecture, chip program, or corporate integration now publicly known. To the extent any of these disclosures would have been material to the filings of other parties, including NASA, the IAU, the AAS, DarkSky International, Amazon/Kuiper, Blue Origin, Viasat, or Petitioner, the pleading cycle was conducted on an incomplete picture that the Applicant had the ability to complete at any time.

J.9. The Visual Record.

SpaceX published, and multiple outlets reproduced, an illustration showing the AI Sat Mini satellite drawn to scale alongside the Starship V3 rocket. The satellite, when deployed with solar arrays and radiators extended, towers over the 124-meter launch vehicle. See SpaceNews (Mar.

22, 2026), reproducing the SpaceX illustration with the caption: "A SpaceX illustration showing the Starship V3 rocket on the same scale as its planned AI Sat Mini orbital data center spacecraft."

The Commission is evaluating an application to authorize up to one million of these objects in low Earth orbit. The certified record before the Commission does not contain the dimensions of a single one.

J.10. Preservation.

Petitioner preserves all arguments arising from the March 21, 2026 disclosures to the fullest extent allowable by law, including their bearing on: the completeness and continuing accuracy of the certified record and the Form 312 certification under 18 U.S.C. Section 1001; the character and candor evaluation under 102 F.C.C.2d 1179; the Section 1.65 ownership amendment obligation; the adequacy of the environmental certification; the environmental analysis obligations under 47 C.F.R. Section 1.1307(c), Executive Order 12114, and Article IX of the Outer Space Treaty; the evidentiary weight of the Opposition's technical representations; the sufficiency of the pleading cycle; the Commission's ability to make the affirmative public interest finding 47 U.S.C. Section 309(a) requires; and the hearing designation threshold under 47 U.S.C. Section 309(e). To the extent the information disclosed at the March 21 event constitutes a material change in the representations underlying the Application, the Applicant's obligation under 47 C.F.R. Section 1.65 to amend is independently triggered, and any supplementation incorporating the disclosed parameters would constitute a major amendment under 47 C.F.R. Section 25.116(b), requiring new public notice and a renewed comment period. To the extent the disclosures bear on the accuracy of the Form 312 certification, the Commission should evaluate whether the certified record remains reconcilable with what the Applicant has now told the world. All arguments in this Appendix are independently maintained for petition for reconsideration under 47 U.S.C. Section 405 and judicial review under 47 U.S.C. Section 402(b) and 5 U.S.C. Section 706.

NOTICE OF DOCUMENT INTEGRITY CERTIFICATION

A Certificate of Document Integrity, including SHA-256 hash values for this Reply and each Appendix filed concurrently herewith, will be filed separately in this docket. That Certificate will confirm the contents of each document as transmitted to the Commission and served on the parties listed above. Any discrepancy between the hash values recorded at time of filing and the documents appearing on the docket is preserved as a record integrity issue for all purposes.

- END OF FILING -